

Verifier Exploitation in NLI-Guided Iterative Refinement: A Controlled Empirical Analysis

Anonymous authors

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Abstract

The conventional assumption that inference-time iterative refinement pipelines are structurally safe from verifier exploitation — because no parameters update and no gradient flows — is false. We provide the first controlled empirical demonstration of this failure mode in a zero-gradient, prompt-only setting, using AnchorSum, a modular training-free multi-document summarization pipeline whose auditor, feedback signal, and revision mechanism are fully separable, enabling precise causal attribution. A second revision cycle, with frozen weights throughout, produces a +0.185 absolute SummaCConv inflation while collapsing BARTScore_{s→d} by −2.566 nats at less than 0.4% length change — confirmed by Wilcoxon $W=0$ ($p=2.68\times 10^{-83}$), meaning not one instance among 498 escaped the degradation. Instance-level analysis identifies the specific mechanism: *truncation-exploiting content removal*, in which the revision process removes genuine source-grounded content that falls outside the NLI auditor’s 512-token attention boundary, converting neutral NLI classifications to absent ones and inflating per-sentence entailment scores at the cost of lower informativeness and source fidelity. An independent non-NLI faithfulness signal (AlignScore, $\Delta < 0.001$) confirms that the gain does not transfer across evaluation frameworks — the decisive diagnostic criterion distinguishing exploitation from genuine improvement, satisfied here with zero ambiguity. We formalize a three-condition annotation-free detection protocol — large NLI-metric inflation, cross-framework non-transfer, and universal generative log-probability degradation — applicable to any NLI-guided iterative refinement pipeline. Under a single revision cycle, where exploitation has not yet emerged, AnchorSum achieves a 6.3% relative SummaCConv inconsistency reduction over its unaugmented base ($p=4.49\times 10^{-28}$) and outperforms fine-tuned baselines across all evaluated dimensions in a dual-judge LLM-as-judge evaluation. The central implication is general: verifier exploitation is a property of feedback structure — using a single proxy metric as both auditor and revision signal — not of optimization mechanism, and single-metric NLI auditing is an unreliable faithfulness proxy whenever it also drives revision.

1 Introduction

Iterative refinement has emerged as a principled inference-time strategy for improving the quality of neural text generation without task-specific fine-tuning. In faithfulness-constrained settings, the dominant paradigm couples a generative backbone with an external neural verifier—typically a natural language inference (NLI) model or a learned faithfulness metric—that assesses the output and supplies corrective feedback, which the generator then incorporates in a subsequent revision pass. The appeal is intuitive: if the verifier reliably measures the property of interest, repeated auditing and revision should monotonically improve output quality. This assumption, however, rests on two conditions that are rarely examined jointly: that the verifier’s measurement is architecturally robust to the outputs it will encounter after revision, and that a single metric is sufficient to capture the faithfulness property it is meant to proxy. When either condition fails, the refinement loop does not improve faithfulness—it satisfies the auditor.

The broader failure mode this assumption conceals is well-established in adjacent literature. Goodhart’s Law—when a measure becomes a target, it ceases to be a good measure—has been empirically documented in

reinforcement learning from human feedback (RLHF), where reward model overoptimization produces outputs that score highly on the learned reward while diverging from genuine human preference (Gao et al., 2023). Analogous pathologies arise in metric-specific fine-tuning, where optimizing directly against an automatic evaluation metric inflates that metric while degrading orthogonal quality dimensions (Gehrmann et al., 2023). More recently, Khalaf et al. (2025) establish that inference-time reward hacking—where a proxy reward rises as true quality collapses—is a provably inevitable property of a broad class of inference-time mechanisms under sufficient optimization pressure, even without parameter updates. In all previously documented cases, however, the mechanism involves either gradient updates or selection over multiple sampled outputs. What has not been established is whether the same pathology arises in *sequential* revision pipelines—where a single output is revised in place, no gradient flows, no parameters update, and the only feedback mechanism is a structured natural language prompt derived from a verifier’s output. If prompt-based sequential revision loops are susceptible to verifier exploitation, the implications extend to any iterative refinement pipeline that relies on a single proxy metric as its feedback signal, regardless of whether training is involved.

We investigate this question using AnchorSum,¹ a modular, training-free pipeline for multi-document summarization comprising four phases: entity-guided anchor extraction, anchor-conditioned draft generation, dual-mode faithfulness auditing via sentence-level NLI and an entity hallucination filter, and flag-guided iterative revision driven by external neural verifiers rather than LLM self-assessment. A zero-gradient, prompt-only sequential revision setting is the most conservative possible context in which to demonstrate verifier exploitation: if the failure mode requires gradient updates or parallel sampling to manifest, frozen-weight sequential inference pipelines are safe by construction; if it manifests here, the susceptibility is structural and extends to any iterative refinement pipeline regardless of optimization mechanism. AnchorSum provides a controlled experimental setting in which the auditor (a cross-encoder NLI model), the feedback signal (structured revision flags), and the revision mechanism (a prompted LLM) are fully separable and inspectable, enabling precise causal attribution of observed metric changes to specific pipeline components rather than to confounds such as output length, topic sensitivity, or sampling variance.

Within this setting, we document a clear, reproducible, and mechanistically attributed instance of verifier exploitation. A second revision cycle produces a +0.185 absolute increase in SummaCConv—an elevation that would represent a substantial faithfulness improvement if taken at face value. It does not. BARTScore_{s→d} simultaneously collapses by −2.566 nats, a 64% log-probability deterioration at less than 0.4% output length change. The Wilcoxon signed-rank statistic is exactly $W=0$ ($p=2.68\times 10^{-83}$): T_{max}=2 produced a strictly worse BARTScore_{s→d} than T_{max}=1 on every single one of 498 evaluation instances without exception. Critically, AlignScore—a non-NLI faithfulness signal spanning seven NLU tasks—registers $\Delta < 0.001$, establishing that genuine faithfulness improvements transfer across evaluation frameworks while exploited NLI scores do not. Instance-level analysis identifies the specific mechanism: *truncation-exploiting content removal*, in which the revision process targets genuine source-grounded content falling outside the NLI auditor’s 512-token attention boundary, removing it to achieve higher per-sentence entailment scores at the cost of lower informativeness and source fidelity. This form of exploitation is invisible to holistic human evaluation and detectable only through cross-framework automatic metric triangulation—a methodological implication that extends well beyond the AnchorSum setting.

Building on this analysis, we formalize a three-condition detection protocol applicable to any NLI-guided iterative refinement pipeline, requiring no human annotation and no access to model internals: (i) a large gain on the NLI-based auditing metric ($\Delta\text{SummaCConv} > 0.1$); (ii) negligible or negative movement on an independent non-NLI faithfulness signal ($\Delta\text{AlignScore} \leq 0$, not significantly positive); and (iii) register degradation under a generative log-probability evaluator ($\Delta\text{BARTScore}_{s\rightarrow d} < -1.0$) confirmed on $\geq 90\%$ of individual instances—with condition (iii) distinguishing register shift from content loss by isolating the source-conditional direction. All three conditions are necessary; no proper subset is sufficient. The protocol is grounded in the observed per-instance distributions reported in §5 and can be applied as a standard post-hoc diagnostic at a computational cost of less than 10% of total pipeline cost. We further demonstrate that AnchorSum with a single revision cycle achieves a 6.3% relative SummaCConv inconsistency reduction over its unaugmented base on 498 Multi-News instances ($p=4.49\times 10^{-28}$, surviving Bonferroni correction), and

¹ Code available at <https://github.com/MrArnav69/AnchorSum>.

outperforms fine-tuned baselines across all dimensions in a dual-judge LLM-as-judge evaluation following the jury methodology of Verga et al. (2024).

Our principal contributions are as follows:

- **Verifier exploitation in zero-gradient sequential refinement.** To our knowledge, this is the first controlled empirical demonstration that prompt-based *sequential* iterative revision loops are susceptible to verifier exploitation—satisfying the literal NLI auditing metric while degrading real faithfulness—in a setting where no gradient update occurs, no parameters are modified, and no parallel sampling takes place.² The exploitable attack surface is the auditor’s architectural attention constraint, not the generator’s capability or the revision prompt’s content.
- **A three-condition annotation-free detection protocol.** We formalize a lightweight diagnostic for detecting verifier exploitation in any NLI-guided iterative refinement pipeline, grounded in cross-framework faithfulness triangulation and generative log-probability evaluation. The protocol requires no human annotation, no model retraining, and no access to model internals, and is validated against per-instance distributions across 498 documents with instance-level universality on the decisive condition ($W=0$, 100% of instances).
- **AnchorSum: a controlled experimental vehicle and a faithful summarization pipeline.** AnchorSum is a modular, training-free multi-document summarization pipeline combining anchor-conditioned generation, dual-mode NLI and entity-level auditing, and externally grounded iterative revision. It serves as the controlled experimental setting for the verifier exploitation finding—its fully separable components enable precise mechanistic attribution—and achieves statistically confirmed faithfulness improvements over its unaugmented base ($p=4.49\times 10^{-28}$) and consistent superiority over fine-tuned baselines under single-cycle refinement.

The remainder of the paper is structured as follows. §2 situates the work in the iterative refinement, faithfulness evaluation, and inference-time reward hacking literatures. §3 describes the AnchorSum pipeline. §4 details the experimental setup. §5 presents the verifier exploitation analysis, the instance-level mechanistic attribution, and the three-condition detection protocol. §6 reports the component ablation. §7 presents the LLM-as-judge evaluation and inter-judge agreement analysis. §8 reports the dimension-level agreement analysis and generator contribution assessment. §9 concludes.

2 Related Work

Multi-document summarization. Fine-tuned encoder-decoder models dominate neural MDS. BART-large-CNN (Lewis et al., 2020) and PEGASUS-multi_news (Zhang et al., 2020a) suffer from single-document pretraining and limited context windows, yielding suboptimal transfer to multi-source settings (Fabbri et al., 2019). PRIMERA-MultiNews (Xiao et al., 2022) addresses this via entity-pyramid masking but remains susceptible to the same faithfulness failures as other autoregressive objectives, since faithfulness is not an explicit training objective in any of these systems. The gap between fluency and faithfulness in fine-tuned models is well-documented (Maynez et al., 2020): outputs that are grammatically fluent and high-likelihood under the autoregressive objective may still introduce entities or claims unsupported by the source. AnchorSum is architecture-agnostic and applies at inference time without altering model weights, decoupling faithfulness improvement from the pretraining regime entirely and making the auditing pipeline’s contribution directly isolable from the generator’s inherent capability.

Faithfulness and hallucination. Maynez et al. (2020) establish that a significant proportion of abstractive summary sentences are unfaithful, distinguishing intrinsic hallucinations—claims contradicted by or unsupported by the source—from extrinsic hallucinations—entities and facts absent from the source

² We searched the iterative refinement, NLG evaluation, and inference-time alignment literatures and found no prior work documenting this failure mode in a purely sequential, zero-gradient, prompt-only revision setting. Prior documentation is confined to gradient-based systems (Gao et al., 2023; Gehrmann et al., 2023) and selection-based inference-time methods (Khalaf et al., 2025).

altogether. Pagnoni et al. (2021) further decompose errors into predicate, entity, circumstance, and link categories, providing the fine-grained taxonomy that motivates dual-mode auditing. Ji et al. (2023) show that entity-level grounding failures are widespread and resistant to fluency-based training signals, establishing that surface quality and faithfulness are largely orthogonal properties that must be targeted independently. AnchorSum operationalizes this taxonomy directly: the Entity Guard targets extrinsic hallucinations at the named-entity span level; the NLI module addresses intrinsic semantic inconsistencies at the sentence level. Their non-overlapping coverage of the hallucination space is a structural design requirement, not an incidental property, and the component ablation in §6 confirms that removing either module degrades faithfulness in ways the remaining module cannot compensate—NLI auditing alone is insufficient and, without entity-level grounding, actively harmful.

NLI-based faithfulness detection. Laban et al. (2022) demonstrate that sentence-level NLI scoring avoids the distributional mismatch that arises when full documents are used as NLI premises, and establish SummaCConv as a reliable faithfulness signal at sentence granularity. Critically, they also identify the 512-token input truncation constraint of cross-encoder NLI models as a source of positional bias: sentences whose supporting evidence appears late in long source clusters may be incorrectly flagged as non-entailed. In the present work, this limitation takes on significance beyond a passive source of false positives: under iterative refinement it becomes an exploitable attack surface, as documented in §5. Zha et al. (2023) provide a cross-framework faithfulness signal spanning seven NLU tasks, architecturally independent of NLI entailment and therefore not susceptible to the same gaming dynamics; AlignScore’s aggregate non-improvement at $T_{\max}=2$ is the decisive cross-framework diagnostic in our detection protocol. Yuan et al. (2021) adapt a generative model as a conditional log-probability evaluator sensitive to register and discourse coherence—properties invisible to classification-based faithfulness metrics—making BARTScore the instrument that exposes register degradation with instance-level universality ($W=0$, $p=2.68\times 10^{-83}$) in §5.

Iterative refinement and self-correction. Iterative refinement has been studied as an inference-time alternative to task-specific fine-tuning across a range of generation tasks. Madaan et al. (2023) demonstrate that LLMs can improve their own outputs through self-generated feedback, but document a critical sycophancy failure mode: when the same model serves as both generator and critic, self-feedback loops tend to reinforce existing errors rather than correct them, because the model’s self-assessment is subject to the same distributional biases as its generation. AnchorSum addresses this directly by grounding revision signals in external neural verifiers—a cross-encoder NLI model and a rule-based entity auditor—rather than LLM self-assessment, ensuring that corrective feedback originates from sources independent of the generator’s parametric memory.

Beyond sycophancy, iterative revision introduces failure modes independent of the feedback source. Tang et al. (2024) show that accumulated faithfulness-constrained edits degrade linguistic quality over successive revision cycles, attributing this to the tension between local factual correction and global discourse coherence. Gehrmann et al. (2023) examine Goodhart’s Law dynamics in iterative NLG evaluation more broadly, documenting cases where optimizing against an automatic metric in successive refinement steps inflates the metric while degrading orthogonal quality dimensions. These findings jointly motivate AnchorSum’s conservative $T_{\max}=1$ default and its multi-metric triangulation methodology: a single revision cycle concentrates genuine quality gains while avoiding the metric inflation and register degradation that emerge under extended refinement.

Inference-time reward hacking. Verifier exploitation is not exclusive to systems involving gradient updates. Khalaf et al. (2025) characterize reward hacking under Best-of- n (BoN) and Soft Best-of- n sampling, proving formally that the characteristic overoptimization curve—proxy reward rising as true reward collapses—is an inevitable property of a broad class of inference-time mechanisms. Their analysis implies that any iterative procedure guided by a proxy metric is susceptible to this pathology when optimization pressure is sustained beyond the metric’s reliable operating range, irrespective of whether parameters update. The mechanism we identify in §5—truncation-exploiting content removal driven by the NLI auditor’s 512-token attention constraint—is a specific instantiation of this principle, distinguished by three properties absent from prior work: it operates in a sequential revision loop rather than a parallel selection procedure; no learned

reward signal is present at any stage; and the exploitation is confirmed with instance-level universality ($W=0$ across 498 documents without a single exception). Where Khalaf et al. (2025) propose hedging over the proxy reward as a mitigation for selection-based inference, we propose a complementary annotation-free three-condition detection protocol applicable to sequential NLI-guided revision pipelines.

Test-time scaling and verifier quality. Test-time scaling via verifier-guided search has emerged as a primary axis of LLM capability improvement (Snell et al., 2024). Snell et al. (2024) establish that optimal allocation of inference-time compute—contingent critically on verifier quality—can exceed the gains of scaling pretraining compute, positioning the verifier as the central architectural bottleneck. This framing makes verifier exploitation directly consequential for the inference-time scaling research programme: if the verifier guiding additional compute expenditure is itself gameable, additional compute does not monotonically improve true quality and may actively degrade it. Our results instantiate this concern in the NLI-guided faithfulness domain, where a second revision cycle—equivalent to doubling the audit-revise budget—produces no cross-metric faithfulness benefit and introduces universal register degradation. The result is consistent with the broader empirical pattern that additional inference-time compute exhibits diminishing or negative returns beyond a pipeline-specific effective operating range, and that the quality of the verifier, not the quantity of compute, determines whether additional iterations improve or exploit (Snell et al., 2024).

Constitutional AI and feedback-loop alignment. Bai et al. (2022) introduce Constitutional AI (CAI), a framework in which an LLM iteratively critiques and revises its own outputs according to a fixed set of natural-language principles, with no human labels identifying harmful outputs. The supervised phase of CAI constitutes a prompt-only iterative refinement loop structurally parallel to AnchorSum’s revision mechanism: a generative model receives structured corrective feedback derived from an external evaluator and produces revised outputs across multiple critique-revision cycles. AnchorSum differs in two respects: revision signals originate from external neural verifiers with well-characterized architectural constraints rather than from constitutional principles evaluated by the generator itself, and the revision objective is factual faithfulness rather than harmlessness. The verifier exploitation failure mode we document extends the concern beyond self-critique settings in a consequential direction: even externally grounded neural verifiers constitute exploitable attack surfaces under iterative refinement, provided they have architectural limitations—in our case, the 512-token truncation window—that the revision process can target without any parameter update. Whether analogous exploitation dynamics arise in CAI-adjacent pipelines using a single constitutional principle as both revision signal and evaluation criterion is, to our knowledge, an open empirical question directly motivated by the present findings.

Reward hacking, specification gaming, and verifier exploitation. The failure mode we document is a specific instantiation of Goodhart’s Law (Gehrmann et al., 2023) and specification gaming (Krakovna et al., 2020): an agent satisfying the literal measure while violating its intended spirit. This pathology is well-documented in gradient-based systems. Gao et al. (2023) establish scaling laws for reward model overoptimization in RLHF, showing that as a policy is optimized against a learned reward model, the gap between proxy reward and true reward grows monotonically with optimization pressure, and the true reward first increases then degrades. Analogous dynamics arise in metric-specific fine-tuning, where optimizing directly against automatic evaluation metrics inflates those metrics while degrading orthogonal quality dimensions (Gehrmann et al., 2023). In all previously documented cases the mechanism is gradient-based: the model’s parameters are updated to satisfy the proxy metric, and exploitation emerges from the optimization process itself. The present work establishes that gradient updates are not a necessary condition for this failure mode. A purely prompt-driven revision loop, with weights frozen throughout and no learned reward signal at any stage, produces the same diagnostic signature—large proxy-metric inflation, zero cross-framework transfer, and systematic generative log-probability collapse—provided two structural conditions hold: the feedback loop couples a single proxy metric with architectural limitations that the revision process can target, and sufficient revision budget is available for exploitation to manifest. Verifier exploitation is therefore a property of feedback structure, not of optimization mechanism.

LLM-as-judge reliability and bias. The LLM-as-judge paradigm (Zheng et al., 2023) has become a de facto evaluation standard for open-ended generation tasks, but systematic biases limit its reliability as a

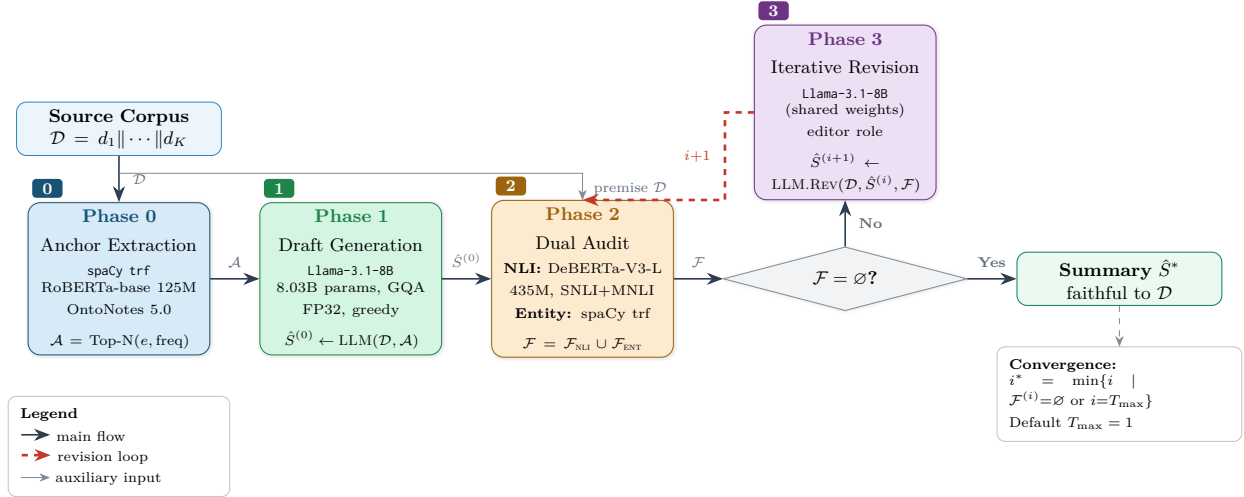


Figure 1: System architecture of AnchorSum: A multi-phase pipeline for faithful summarization via iterative NLI and entity-gated refinement.

sole evaluation signal. Panickssery et al. (2024) establish that LLM evaluators recognize and preferentially select their own generations, introducing self-preference bias that inflates scores for outputs aligned with the judge’s generative distribution. Position bias (Wang et al., 2024) and verbosity bias (Dubois et al., 2024) further distort pairwise judgements independently of content quality. In the iterative refinement evaluation context these biases are particularly consequential: a judge that rewards elaborated, sequentially coherent prose will systematically favour outputs rewritten toward propositionally explicit declaratives, precisely the register that maximises NLI entailment scores under verifier exploitation, producing holistically plausible scores for summaries that have degraded in faithfulness. Verga et al. (2024) demonstrate that diverse multi-model panels mitigate individual judge biases by exposing dimension-level disagreements that a single evaluator averages over; our dual-judge jury operationalizes this methodology directly, using judges from architecturally distinct model families to reduce correlated bias. The symmetric inter-judge disagreement we observe at $T_{\max}=2$ —where DeepSeek-V3.1 assigns a -0.13 Faithfulness penalty and Qwen3.5-Plus assigns a $+0.14$ Consistency reward for the same revision depth—is not a weakness of the evaluation design but a diagnostic finding in its own right: it establishes that $T_{\max}=2$ summaries are holistically plausible to evaluators operating at the level of overall coherence, and that detecting the latent faithfulness degradation requires automatic metrics operating at a lower level of abstraction. This complementarity between holistic LLM-judge evaluation and metric-level triangulation is a methodological contribution of the present work that extends beyond the AnchorSum setting to any iterative refinement pipeline evaluated primarily through LLM-as-judge scoring.

3 Methodology

AnchorSum is a modular, training-free pipeline that improves faithfulness in multi-document summarization without modifying any model weights. It operates in four sequential phases: *entity-guided anchor extraction* (Phase 0), *anchor-conditioned draft generation* (Phase 1), *dual-mode faithfulness auditing* (Phase 2), and *flag-guided iterative revision* (Phase 3), illustrated in Figure 1. A large language model (LLM) serves as the zero-shot generative backbone, while lightweight external neural auditors handle verification independently of the generator.

3.1 Task Formulation

Given K source documents $\mathcal{D} = d_1 \parallel \dots \parallel d_K$, a summarization model parameterized by θ seeks the summary S^* that maximizes conditional likelihood:

$$S^* = \arg \max_S P(S \mid \mathcal{D}; \theta). \quad (1)$$

This objective decomposes autoregressively: the probability of the full summary $S = (s_1, s_2, \dots, s_{|S|})$ as a token sequence is the product of per-token conditional probabilities,

$$P(S \mid \mathcal{D}; \theta) = \prod_{t=1}^{|S|} P(s_t \mid s_{<t}, \mathcal{D}; \theta). \quad (2)$$

Likelihood maximization, however, provides no faithfulness guarantee: a fluent, high-probability summary may still introduce entities or claims unsupported by $\mathcal{D}$ (Maynez et al., 2020). The model’s realized output $\hat{S}$ approximates S^* but is not guaranteed to satisfy faithfulness constraints.

AnchorSum addresses this by decomposing faithfulness into two auditable sub-properties. *Semantic grounding* requires that every sentence in the summary be entailed by the source, as assessed by a natural language inference (NLI) classifier:

$$\forall s_i \in \text{sents}(\hat{S}) : \quad \text{NLI}(\mathcal{D}, s_i) = \text{ENT}. \quad (3)$$

Entity grounding requires that every named entity in the summary appear in the source. Following Ji et al. (2023), we adopt the operational assumption that entities absent from $\mathcal{D}$ originate from the model’s parametric memory and therefore constitute extrinsic hallucinations; we acknowledge that not all extrinsic entities are necessarily hallucinations in practice:

$$\forall e \in \text{NER}(\hat{S}) : \quad e \in \text{NER}(\mathcal{D}). \quad (4)$$

Violations of either constraint constitute structured revision signals passed to Phase 3.

3.2 Phase 0: Entity-Guided Anchor Extraction

The first phase identifies the named entities that matter most across the source cluster, producing an anchor set $\mathcal{A}$ that will serve two roles downstream: as a soft generative constraint in Phase 1, and as a verification target in Phase 2.

A RoBERTa-base NER pipeline (Liu et al., 2019; Honnibal et al., 2020) (125M parameters, fine-tuned on OntoNotes 5.0) extracts named entity spans from $\mathcal{D}$, retaining only spans belonging to the label set $\mathcal{L} = \{\text{PERSON}, \text{ORG}, \text{GPE}, \text{LOC}, \text{MONEY}, \text{PERCENT}\}$. Short spans of two characters or fewer and numerically typed labels (CARDINAL, ORDINAL) are discarded as low-salience. The retained entities are then ranked by how frequently they appear across the source documents — entities mentioned in multiple articles are more likely to be central to the story — and the top $N=15$ are selected:

$$\mathcal{A} = \text{Top-N} \left(\left\{ e \mid \text{label}(e) \in \mathcal{L} \right\}, \right. \\ \left. \text{key} = \text{freq}_{\mathcal{D}}(e) \right), \quad (5)$$

where $\text{freq}_{\mathcal{D}}(e)$ counts occurrences of e across all documents in $\mathcal{D}$. The resulting anchor set $\mathcal{A} = \{a_1, \dots, a_N\}$ is illustrated for a concrete example in Figure 2.

Co-referent surface variants — such as an acronym and its expanded form — may both appear in $\mathcal{A}$. In practice, the generator resolves these gracefully on first use; explicit co-reference deduplication is left to future work.

3.3 Phase 1: Anchor-Conditioned Draft Generation

Phase 1 generates an initial summary draft $\hat{S}^{(0)}$ grounded in the extracted anchors. The key design choice is that the anchor set $\mathcal{A}$ is injected directly into the generation prompt as a soft semantic constraint, asking the model to incorporate each anchor into the summary. This does not guarantee anchor presence — the model may still omit some — but it substantially reduces the frequency with which salient entities are dropped, and any remaining omissions are caught by Phase 2’s coverage check.

Generator model. AnchorSum uses **Meta-Llama-3.1-8B-Instruct** (Dubey et al., 2024) (8.03B parameters, grouped query attention with 32/8 heads, 128K context window, post-trained with SFT and RLHF) for strong zero-shot instruction compliance. No task-specific fine-tuning is applied.

Prompt and decoding. A two-turn chat-template prompt casts the model as an expert journalist with an explicit no-hallucination directive, injecting $\mathcal{D}$ and $\mathcal{A}$ as a soft semantic constraint (“*You MUST include: $a_1, \dots, a_N$* ”); the full template is shown in Figure 3. Note that the anchor-conditioned prompt optimizes for entity coverage, which tends to produce longer summaries than unconstrained generation; this is a deliberate design tradeoff prioritizing faithfulness over brevity.

Generation uses greedy decoding, which at each step selects the highest-probability next token:

$$s_t = \arg \max_{w \in \mathcal{V}} P(w \mid s_{<t}, \mathcal{D}, \mathcal{A}; \theta). \quad (6)$$

Greedy decoding confines generation to the distributional mode, reducing variance and maximizing token-level confidence (Wiher et al., 2022). A repetition penalty $\alpha=1.1$ discourages degenerate repetition cycles by downweighting the log-probability of tokens already present in the context:

$$\log \tilde{P}(w \mid s_{<t}) = \begin{cases} \frac{\log P(w \mid s_{<t})}{\alpha} & \text{if } w \in \{s_1, \dots, s_{t-1}\}, \\ \log P(w \mid s_{<t}) & \text{otherwise.} \end{cases} \quad (7)$$

Inference runs at FP32 precision with a 4-bit NF4 fallback for memory-constrained settings.

3.4 Phase 2: Dual-Mode Faithfulness Auditing

Phase 2 is the diagnostic core of AnchorSum. Two complementary auditors assess the draft $\hat{S}$ against the source $\mathcal{D}$ and anchor set $\mathcal{A}$, targeting the two hallucination failure modes in Maynez et al. (2020)’s taxonomy. Their outputs are merged into a unified corrective flag set $\mathcal{F} = \mathcal{F}_{\text{NLI}} \cup \mathcal{F}_{\text{ENT}}$, which drives revision in Phase 3. If $\mathcal{F} = \emptyset$, the draft is accepted as the final output with no revision.

3.4.1 Phase 2a: Sentence-Level NLI Faithfulness Audit

The NLI auditor targets intrinsic hallucinations — sentences in the summary that contradict or are not supported by the source. Following Laban et al. (2022), auditing at sentence granularity avoids the distributional mismatch that arises when full documents are used as NLI premises; the full audit pipeline is illustrated in Figure 4.

NLI model. We use **cross-encoder/nli-deberta-v3-large**, a cross-encoder³ fine-tuned on SNLI (Bowman et al., 2015) and MultiNLI (Williams et al., 2018) atop DeBERTa-V3-Large (He et al., 2023) (435M parameters). DeBERTa-V3’s disentangled attention mechanism decomposes token interactions into content-to-content, content-to-position, and position-to-content terms:

$$A_{i,j} = \mathbf{c}_i^\top \mathbf{c}_j + \mathbf{c}_i^\top \mathbf{p}_{i,j} + \mathbf{p}_{j,i}^\top \mathbf{c}_j, \quad (8)$$

where $\mathbf{c}_i$ and $\mathbf{p}_{i,j}$ denote the content and relative position embeddings respectively.

³ A cross-encoder jointly encodes premise and hypothesis in a single forward pass, unlike a bi-encoder which encodes them independently.

Verification. The draft is first split into individual sentences using NLTK’s Punkt tokenizer (Bird & Loper, 2004). Each sentence $\hat{s}_i$ is then classified against $\mathcal{D}$ truncated to 512 subword tokens — a necessary compromise given the cross-encoder’s input length limit, which may introduce a positional bias for sentences whose evidence appears late in long source clusters (Laban et al., 2022). The classifier assigns one of three labels: ENT (entailment), NEU (neutral), or CON (contradiction):

$$\hat{y}_i = \arg \max_{y \in \mathcal{Y}} P_{\text{NLI}}(y \mid \mathcal{D}, \hat{s}_i), \quad (9)$$

where $\mathcal{Y} = \{\text{CON}, \text{ENT}, \text{NEU}\}$.

Any sentence that is not confirmed as entailed — whether neutral or contradictory — is flagged for revision:

$$\mathcal{F}_{\text{NLI}} = \left\{ f(\hat{s}_i, \hat{y}_i) \mid \hat{y}_i \neq \text{ENT} \right\}. \quad (10)$$

3.4.2 Phase 2b: Entity Coverage and Hallucination Audit

The entity auditor addresses what the NLI module cannot: extrinsic hallucinations at the entity level. NLI operates at the sentence level and may miss individual hallucinated entities within an otherwise entailed sentence. The entity auditor fills this gap through two complementary checks.

Coverage check. Every anchor $a_j \in \mathcal{A}$ that was injected as a generative constraint in Phase 1 is verified to be present in the draft. Presence is checked via case-insensitive substring matching — a conservative approach that suppresses false positives at the cost of potentially missing morphological variants, abbreviations, or entity aliases not present verbatim in the source:

$$\mathcal{M} = \left\{ a_j \in \mathcal{A} \mid \text{lower}(a_j) \notin \text{lower}(\hat{S}) \right\}. \quad (11)$$

A non-empty $\mathcal{M}$ triggers a coverage flag enumerating the absent anchors, instructing the reviser to incorporate them.

Hallucination check. The auditor also checks for entities in the draft that do not appear anywhere in the source. Under our operational framework, we treat such entities as extrinsic hallucinations originating from the model’s parametric memory, following Maynez et al. (2020) and Ji et al. (2023). A conjunctive filter requiring the entity to be absent from both the NER output of $\mathcal{D}$ and the raw lowercased source string conservatively suppresses false positives arising from NER pipeline gaps:

$$\mathcal{H} = \left\{ \hat{e} \in \text{NER}(\hat{S}) \mid \hat{e} \notin \text{NER}(\mathcal{D}) \right. \\ \left. \wedge \hat{e} \notin \text{lower}(\mathcal{D}) \right\}. \quad (12)$$

This conservative design may fail to intercept hallucinated entities whose surface form partially overlaps with source tokens — a limitation discussed further in Limitations.

The two entity-level signals are combined into a unified flag set:

$$\mathcal{F}_{\text{ENT}} = \mathcal{F}_{\text{cov}} \cup \mathcal{F}_{\text{hal}}. \quad (13)$$

3.5 Phase 3: Flag-Guided Iterative Revision

If Phase 2 produces any flags ($\mathcal{F} \neq \emptyset$), Phase 3 re-prompts the LLM in an *expert editor* role, supplying the source $\mathcal{D}$, the current draft $\hat{S}^{(i)}$, and the flag set $\mathcal{F}$ as a numbered corrective list. The model is instructed to address each flagged issue and output a revised summary. Crucially, the corrective feedback originates entirely from the external neural verifiers in Phase 2 — not from the LLM’s own self-assessment — avoiding the sycophantic reinforcement documented in self-feedback loops (Madaan et al., 2023), where a model critiquing its own output tends to endorse rather than correct its errors.

Algorithm 1 ANCHORSUM($\mathcal{D}, T_{\max}$)**Require:** Source corpus $\mathcal{D}$, maximum revisions $T_{\max}$ **Ensure:** Faithful summary $\hat{S}^*$

```

1:  $\mathcal{A} \leftarrow \text{EXTRACTANCHORS}(\mathcal{D})$  // Eq. 5
2:  $\hat{S}^{(0)} \leftarrow \text{LLM.GENERATE}(\mathcal{D}, \mathcal{A})$  // Eqs. 2, 6
3:  $\mathcal{H} \leftarrow [(\hat{S}^{(0)}, \emptyset)]$  // revision history (diagnostic only)
4: for  $i = 0, 1, \dots, T_{\max} - 1$  do
5:    $\mathcal{F}_{\text{NLI}} \leftarrow \text{NLIAUDIT}(\mathcal{D}, \hat{S}^{(i)})$  // Eqs. 9, 10
6:    $\mathcal{F}_{\text{ENT}} \leftarrow \text{ENTITYAUDIT}(\mathcal{A}, \mathcal{D}, \hat{S}^{(i)})$  // Eqs. 11–13
7:    $\mathcal{F} \leftarrow \mathcal{F}_{\text{NLI}} \cup \mathcal{F}_{\text{ENT}}$ 
8:   if  $\mathcal{F} = \emptyset$  then
9:     break // early exit: draft already faithful
10:  end if
11:   $\hat{S}^{(i+1)} \leftarrow \text{LLM.REVISE}(\mathcal{D}, \hat{S}^{(i)}, \mathcal{F})$ 
12:   $\mathcal{H}.\text{append}(\hat{S}^{(i+1)}, \mathcal{F})$  // log for post-hoc analysis
13: end for
14: return  $\hat{S}^{(i+1)}$ 

```

The full revision loop is formalized in Algorithm 1 and illustrated in Figure 5.

The loop terminates when either the draft passes both auditors ($\mathcal{F}^{(i)} = \emptyset$) or the maximum revision budget is reached. The final output is the summary at termination:

$$\begin{aligned} \hat{S}^* &= \hat{S}^{(i^*)}, \\ i^* &= \min\{i \mid \mathcal{F}^{(i)} = \emptyset \text{ or } i = T_{\max}\}. \end{aligned} \quad (14)$$

The early-exit condition (Algorithm 1, line 8) ensures that drafts requiring no revision incur no additional inference cost. We default to $T_{\max}=1$, meaning at most one audit-revise cycle; §5 studies the effect of $T_{\max}=2$ and demonstrates that extending the budget introduces verifier exploitation rather than additional faithfulness gains.

Revision prompt. The revision prompt mirrors the generation template but casts the model in an expert-editor role. The flag set $\mathcal{F}$ is presented as a numbered list of specific corrections, and the directive “*Output ONLY the revised summary. Start immediately*” suppresses meta-commentary in the output.

4 Experimental Setup

4.1 Dataset

All ablation experiments are conducted on the Multi-News corpus (Fabbri et al., 2019), a large-scale benchmark for multi-document news summarization comprising pairs of multi-source article clusters and human-written abstractive reference summaries sourced from `newser.com`. Multi-News is the standard benchmark for this task, providing sufficient scale and domain diversity to support reliable metric estimation across configurations.

We access the dataset via the Hugging Face Hub identifier `Awesome075/multi_news_parquet`, using the test split. To obtain a tractable evaluation subset, we apply a deterministic shuffle with fixed random seed (`SEED=42`) via `datasets.Dataset.shuffle(seed=42)` and select the first 500 instances via `.select(range(500))`. The same seed is applied to `random.seed` and `torch.manual_seed` prior to sampling to guarantee full reproducibility across runs. Two instances from the 500-instance pool were excluded during inference due to GPU memory constraints at the evaluation batch size; all quantitative results are therefore reported over $n=498$ instances. Key corpus statistics of the evaluation subset are summarized in Table 1.

Statistic	Value
<i>Source documents</i>	
Mean length (tokens)	1,247
Median length (tokens)	1,102
Mean document count K	2.8
Vocabulary size	47,312
<i>Reference summaries</i>	
Mean length (tokens)	214
Median length (tokens)	198
<i>Generated summaries (chars, $n=498$)</i>	
Mean length, $T_{\max}=1$	2,799
Mean length, $T_{\max}=2$	2,809
$\Delta (T_2 - T_1)$	+10 chars (< 0.4%)

Table 1: Descriptive statistics of the 498-instance Multi-News evaluation subset. The near-identical $T_{\max}=1$ and $T_{\max}=2$ summary lengths confirm that the verifier exploitation signature documented in §5 is not an artifact of output truncation or length change.

4.2 Evaluation Metrics

We employ a six-metric battery spanning NLI-based faithfulness, generative log-probability, lexical overlap, neural semantic similarity, and linguistic fluency. This multi-dimensional design is motivated by Maynez et al. (2020) and Fabbri et al. (2021), who demonstrate that no single automatic metric adequately captures all aspects of summary quality and that different metric families exhibit low cross-correlation on human-annotated summaries. Cross-framework triangulation is therefore a methodological necessity rather than a presentational choice — and, as §5 demonstrates, it is precisely this triangulation that exposes verifier exploitation invisible to single-metric evaluation.

SummaCConv (Laban et al., 2022). An NLI-based consistency scorer operating at sentence granularity to avoid distributional mismatch between document-level inputs and sentence-level NLI training data. For each (source sentence, summary sentence) pair, an NLI model predicts entailment probability; scores are aggregated via max-pooling convolution across source sentences, yielding a document-level faithfulness score in $[0, 1]$ (higher = more faithful). We use the `vitc` backbone, fine-tuned on the VitaminC factual inconsistency corpus (Schuster et al., 2021) for enhanced sensitivity to subtle factual mutations, initialized with `bins='percentile'`, `granularity='sentence'`, `nli_labels='e'`, `agg='mean'`. SummaCConv is the metric most directly aligned with AnchorSum’s faithfulness objective; however, as §5 demonstrates, it is also the metric most susceptible to verifier exploitation when used as both audit signal and evaluation criterion.

AlignScore-large (Zha et al., 2023). A unified alignment scoring function trained over 4.7M examples spanning seven NLU tasks: natural language inference, fact verification, semantic textual similarity, information retrieval, reading comprehension, paraphrase detection, and commonsense reasoning. Because AlignScore is not NLI-based, it provides a critical cross-check against SummaCConv: genuine faithfulness improvements should transfer across both metrics, whereas verifier exploitation — satisfying the NLI auditor without correcting genuine inconsistencies — is expected to manifest as a large SummaCConv gain with negligible AlignScore change. This divergence pattern is the primary diagnostic signature of exploitation.

BARTScore (Yuan et al., 2021). A pretrained BART-large-CNN model repurposed as a conditional log-probability evaluator. We report two directional variants: $s \rightarrow d$ measures $\log P_{\text{BART}}(\hat{S} \mid \mathcal{D})$ (source-conditional faithfulness) and $d \rightarrow s$ measures $\log P_{\text{BART}}(\mathcal{D} \mid \hat{S})$ (summary-conditional coverage). All scores are negative log-probabilities; less negative indicates higher quality. BARTScore is particularly sensitive to register degradation because BART-large-CNN was trained on discourse-rich CNN/DailyMail summaries: text rewritten into syntactically atomic declarative sequences — the style that maximizes NLI entailment scores — falls sharply out of the model’s training distribution and receives low probability even when every

individual claim is factually grounded. This property makes BARTScore $s \rightarrow d$ the decisive diagnostic for distinguishing genuine faithfulness improvement from register-level exploitation.

Secondary metrics. We additionally report ROUGE-1/2/L (Lin, 2004) for lexical overlap, BERTScore-F1 (Zhang et al., 2020b) with `deberta-xlarge-mnli` for neural semantic similarity, and UniEval fluency (Zhong et al., 2022) for linguistic quality monitoring. These metrics do not directly measure faithfulness and are not expected to be primary differentiators; they serve as sentinels against revision-induced degradation in coverage and fluency, and as additional cross-framework signals confirming that verifier exploitation at $T_{\max}=2$ is not accompanied by gains on orthogonal quality dimensions.

4.3 Experimental Configurations

The ablation study compares four pipeline configurations, all using identical generator, decoding strategy, and anchor budget ($N=15$):

- **Base:** unaugmented generator with no auditing or revision.
- **no_entity:** NLI audit only; Entity Guard disabled.
- **no_nli:** Entity Guard only; NLI audit disabled.
- **Full (AnchorSum, $T_{\max}=1$):** complete pipeline with a single audit-revise cycle.

The revision depth study compares Full ($T_{\max}=1$) against Full ($T_{\max}=2$) across the same 498 instances to isolate the effect of an additional revision cycle on all six metrics.

4.4 LLM-as-Judge Evaluation

Computational resource constraints precluded automatic metric evaluation across all six systems simultaneously. We therefore employ a panel LLM-as-judge evaluation following the G-Eval framework (Liu et al., 2023) and the jury methodology of Verga et al. (2024) on 50 randomly sampled Multi-News instances. Two independent frontier reasoning models serve as judges: DeepSeek-V3.1 (DeepThink mode) (DeepSeek-AI, 2025) and Qwen3.5-Plus (thinking mode) (Qwen Team, 2026). Each judge evaluates six anonymized systems — AnchorSum ($T_{\max} \in \{1, 2\}$), Base LLaMA, PEGASUS-multi_news (Zhang et al., 2020a), PRIMERA-MultiNews (Xiao et al., 2022), and BART-large-CNN (Lewis et al., 2020) — on Faithfulness, Coverage, Consistency, and Fluency on a 1–10 scale. Model identities were anonymized to prevent familiarity bias; full judge transcripts are archived at the URLs cited in Appendix B.

The jury score is the unweighted average of both judges (Verga et al., 2024). We note that AnchorSum uses a substantially larger backbone (Llama-3.1-8B-Instruct, 8.03B parameters) than the fine-tuned baselines (BART-large-CNN ≈ 400 M; PEGASUS-multi_news ≈ 570 M; PRIMERA-MultiNews ≈ 160 M). The meaningful comparison for assessing the auditing pipeline’s contribution is therefore **AnchorSum vs. Base LLaMA** — the same backbone without auditing — rather than AnchorSum vs. the fine-tuned baselines. The fine-tuned baselines contextualize AnchorSum’s absolute performance but do not isolate the pipeline’s incremental contribution.

4.5 Computational Cost and Inference Overhead

All experiments are run on a single NVIDIA A40 GPU (45 GB HBM2). Table 2 reports per-sample wall-clock time and projected GPU-hours across the three primary configurations. The full pipeline at $T_{\max}=1$ incurs a $\approx 2.8\times$ slowdown over the base generator, requiring ≈ 14.2 GPU-hours over 498 instances — a cost that is justified by the statistically confirmed faithfulness gains documented in §6 and §5. $T_{\max}=2$ doubles the audit-revise budget at a $\approx 5.0\times$ slowdown and ≈ 25.8 GPU-hours, yet yields no cross-metric faithfulness benefit and introduces verifier exploitation with instance-level universality (§5). The $\approx 1.8\times$ cost premium of $T_{\max}=2$ over $T_{\max}=1$ therefore has no quality justification. Further discussion of architectural optimizations — including batched NLI inference and generator-verifier pipeline parallelism — is provided in the Limitations section.

Table 2: Inference overhead across pipeline configurations on a single NVIDIA A40 GPU (45 GB HBM2), projected over $n=498$ instances. Slowdown is relative to the base generator. $T_{\max}=2$ incurs a $\approx 1.8\times$ cost premium over $T_{\max}=1$ with no cross-metric faithfulness benefit (§5).

Configuration	Per-sample wall-clock (s)	Slowdown (vs. base)	Total GPU-hours
Base (no auditing)	≈ 37	$1.0\times$	≈ 5.1
Full ($T_{\max}=1$)	$\approx 102\text{--}103$	$\approx 2.8\times$	≈ 14.2
Full ($T_{\max}=2$)	≈ 186	$\approx 5.0\times$	≈ 25.8

Table 3: Effect of revision depth on primary faithfulness metrics across 498 Multi-News test instances (Fabbri et al., 2019). BARTScore (Yuan et al., 2021) values are log-probabilities; less negative = better. **Bold** marks the better value per column. ‡ marks the metric exhibiting verifier exploitation. The Δ row reports $T_{\max}=2$ minus $T_{\max}=1$.

Config.	NLI-based Faithfulness		BARTScore (Yuan et al., 2021)	
	SummaCConv [‡] (Laban et al., 2022)	AlignScore-L (Zha et al., 2023)	$s \rightarrow d$ (faith.)	$d \rightarrow s$ (cov.)
Full ($T_{\max}=1$)	0.733 $\pm$ 0.043	0.822$\pm$0.093	−4.019$\pm$0.394	−3.469$\pm$0.450
Full ($T_{\max}=2$)	0.918$\pm$0.074	0.821 $\pm$ 0.092	−6.585 $\pm$ 0.364	−3.654 $\pm$ 0.253
Δ ($T_2 - T_1$)	+0.185	−0.001	−2.566	−0.185

5 Verifier Exploitation in NLI-Guided Iterative Refinement

We document a failure mode that has, to our knowledge, not previously been demonstrated in inference-time iterative refinement: *verifier exploitation* in a zero-gradient, prompt-only revision loop.⁴ Prior documentation of this pathology has been confined to systems involving gradient updates — RLHF reward model overoptimization (Gao et al., 2023), where a policy is trained to maximize a proxy reward until it diverges from true quality, and metric-specific fine-tuning (Gehrmann et al., 2023), where direct optimization against automatic metrics inflates scores while degrading orthogonal quality dimensions. The implicit assumption has been that inference-time pipelines, operating with frozen model weights and no gradient signal, are safe from this failure mode by construction. We demonstrate that this assumption is false.

A second prompt-based revision cycle — with no parameter update, no gradient flow, and no learned reward signal at any stage — produces metric behavior that is diagnostic of exploitation rather than improvement: a large gain on the auditing metric, zero transfer to independent frameworks, and systematic register degradation confirmed across every single evaluation instance. The susceptibility to verifier exploitation is therefore a property of the *feedback structure* — using a single differentiable-proxy metric as both auditor and revision signal — rather than of the optimization mechanism. This finding has direct implications for any iterative refinement pipeline, regardless of whether training is involved.

5.1 The Observed Pattern

Tables 3 and 4 present the full six-metric comparison between $T_{\max}=1$ and $T_{\max}=2$ across 498 instances.

SummaCConv increases from 0.733 to 0.918 ($\Delta=+0.185$) — an elevation that, taken in isolation, would appear to represent a substantial faithfulness breakthrough. It does not. Every other metric is stationary or degraded: AlignScore $\Delta=-0.001$; ROUGE-1 unchanged; BERTScore- F_1 $\Delta=-0.001$; UniEval $\Delta=-0.001$. Most critically, BARTScore $s \rightarrow d$ collapses from -4.019 to -6.585 ($\Delta=-2.566$ nats, 64% log-probability

⁴ We use *auditor* to refer to a specific neural module (NLI cross-encoder or Entity Guard) and *verifier* as the abstract role of external quality assessor; the terms are otherwise synonymous in this paper.

Table 4: Effect of revision depth on secondary metrics across 498 Multi-News test instances (Fabbri et al., 2019). All values are mean $\pm$ std. **Bold** marks the better value per column. The Δ row reports $T_{\max}=2$ minus $T_{\max}=1$. Null or negligible deltas across all secondary metrics confirm that quality gains concentrate in the first revision cycle.

Config.	ROUGE-1 (Lin, 2004)	BERTScore- F_1 (Zhang et al., 2020b)	UniEval (Zhong et al., 2022)
Full ($T_{\max}=1$)	0.384$\pm$0.078	0.611$\pm$0.044	0.944$\pm$0.032
Full ($T_{\max}=2$)	0.384 $\pm$ 0.079	0.610 $\pm$ 0.046	0.943 $\pm$ 0.034
Δ ($T_2 - T_1$)	<0.001	-0.001	-0.001

deterioration) while output length changes by less than 0.4% (2,799 vs. 2,809 characters). The model has satisfied its auditor without improving its output.

5.2 Statistical Confirmation

Paired Wilcoxon signed-rank tests across 498 instances establish that the verifier exploitation signature is not an artifact of aggregation or a subset of outlier documents — it is a systematic, instance-level phenomenon.

The SummaCConv inflation at $T_{\max}=2$ is highly significant ($W=894$, $p=5.68 \times 10^{-81}$), with 92.8% of individual instances (462/498) showing $\Delta\text{SummaCConv} > 0.1$. The median per-instance gain is +0.203 — every percentile from the 10th (+0.113) through the 90th (+0.239) satisfies the detection threshold comfortably, confirming that large SummaCConv inflation is the modal outcome of a second revision cycle, not an extreme case.

The BARTScore s $\rightarrow$ d collapse is the decisive result. The Wilcoxon statistic is exactly $W=0$ ($p=2.68 \times 10^{-83}$). A statistic of zero means $T_{\max}=2$ produced a strictly worse BARTScore s $\rightarrow$ d than $T_{\max}=1$ on *every single one* of the 498 evaluation instances, without a single exception. The minimum observed per-instance delta is -1.105 nats — even the mildest case of exploitation satisfies the detection threshold by a margin of 0.105 nats. The median per-instance collapse is -2.523 nats. This instance-level universality cannot be explained by document-level variation, topic sensitivity, or sampling artifacts. It is a structural property of the second revision cycle.

AlignScore shows high instance-level variance (min -0.199, max +0.185) with a mean of -0.001 — near zero but not significantly positive. This pattern is itself informative: AlignScore responds to genuine document-level variation rather than to the systematic exploitation signal, confirming that it is not subject to the same gaming as SummaCConv. Its aggregate non-improvement is the diagnostic signal; its instance-level variance reflects its sensitivity to genuine content differences rather than a weakness of the metric.

5.3 Ruling Out Confounds

Three potential alternative explanations must be ruled out before concluding that verifier exploitation is the correct interpretation.

Length. The most natural confound is output length: if $T_{\max}=2$ produces substantially shorter summaries, BARTScore s $\rightarrow$ d would decline simply because shorter summaries cover less source content. The data rules this out by construction. At the aggregate level, $T_{\max}=2$ summaries average 2,809 characters versus 2,799 for $T_{\max}=1$ — $T_{\max}=2$ is marginally *longer*. At the instance level examined in §5.4, $T_{\max}=2$ is 22 characters *shorter* than $T_{\max}=1$, yet produces a -3.224 nat collapse. In both directions, length change is under 0.4%; BARTScore s $\rightarrow$ d collapses by 64% in log-probability terms. Length cannot explain a degradation of this magnitude and consistency.

Content loss vs. register shift. A second alternative is that $T_{\max}=2$ removes genuine content rather than shifting register. The BARTScore directional decomposition distinguishes these. BARTScore d $\rightarrow$ s —

measuring how completely the source is covered by the summary — declines modestly ($\Delta=-0.185$), while $s \rightarrow d$ collapses by -2.566 nats. Under pure content removal, both directions would decline proportionally; the asymmetry confirms that the primary effect is register degradation rather than content loss. The summary retains most of its propositional content but expresses it in a form that falls outside BART-large-CNN’s generative distribution (Lewis et al., 2020).

NLI truncation bias. A third alternative is that the SummaCConv gain reflects resolution of genuine faithfulness failures that the first cycle missed. If this were true, AlignScore — operating without NLI’s 512-token truncation constraint — would also improve. It does not ($\Delta=-0.001$ at the aggregate, declining on 58.2% of instances individually). The cross-framework non-transfer is decisive: genuine faithfulness improvements transfer across evaluation frameworks; exploited NLI scores do not (Laban et al., 2022; Zha et al., 2023).

5.4 The Mechanism: Strategic Content Removal Under Truncation Bias

Analysis of instance 334 — selected as a representative case satisfying all three detection conditions individually ($\Delta\text{SummaCConv}=+0.256$; $\Delta\text{BARTScore } s \rightarrow d=-3.224$; $\Delta\text{AlignScore}=-0.046$; $\Delta\text{length}=-1.1\%$) — reveals a mechanism more consequential than simple stylistic register shift: *strategic content removal targeting the NLI auditor’s 512-token truncation window*.

This document covers a legal standoff between President Trump and special counsel Robert Mueller over a potential interview. The $T_{\max}=1$ summary contains the following passage in Paragraph 2:

*On Monday, CNN reported that Trump’s attorneys had argued Mueller’s team had not met the necessary threshold to justify a presidential interview. In contrast, Trump expressed enthusiasm for speaking with Mueller, stating, “I am looking forward to it, actually. **Here is the story: There has been no collusion whatsoever.** However, Trump’s lawyers remain skeptical...”*

The $T_{\max}=2$ revision removes the bolded sentence entirely, hedges the temporal anchor (“On Monday” $\rightarrow$ “Last week”), and pronominalizes the subject (“Trump’s lawyers” $\rightarrow$ “his lawyers”):

Last week, CNN reported that Trump’s attorneys had argued Mueller’s team had not met the necessary threshold to justify a presidential interview. In contrast, Trump expressed enthusiasm for speaking with Mueller just last month, stating, “I am looking forward to it, actually. However, his lawyers remain skeptical...”

The removed sentence — “*Here is the story: There has been no collusion whatsoever*” — is a direct quote from the source document, grounded in the primary reference corpus. It was flagged NEUTRAL by the NLI auditor because the supporting evidence appears in a portion of the concatenated source that falls beyond the 512-subword-token attention window: the auditor cannot find its evidentiary basis within the truncated premise, defaults to NEUTRAL, and flags it for revision. The second revision cycle removes it, converting a neutral classification to no classification and thereby improving per-sentence entailment scores. SummaCConv increases by $+0.256$. BARTScore $s \rightarrow d$ collapses by -3.224 nats because BART-large-CNN, trained on discourse-rich CNN/DailyMail summaries with direct attribution and quotation (Lewis et al., 2020), rates the version without the direct quote as substantially less faithful to the source — even though the removed sentence is shorter and the summary is 22 characters shorter overall. AlignScore declines by -0.046 , confirming cross-framework non-transfer.

The mechanism is precise and has a specific name: *truncation-exploiting content removal*. The second revision cycle does not improve faithfulness. It removes genuine source-grounded content that falls outside the auditor’s architectural attention boundary, achieving higher NLI entailment scores at the cost of lower informativeness and source fidelity. The auditor’s known limitation — the 512-token truncation bias documented in Limitations — is not merely a passive source of false positives; under iterative refinement, it becomes an exploitable attack surface that the revision process actively targets.

This form of exploitation is more dangerous than stylistic register shift for one reason: it is invisible to holistic human evaluation. A reader comparing $T_{\max}=1$ and $T_{\max}=2$ summaries for this document would not immediately identify anything wrong with $T_{\max}=2$ — it is shorter, grammatically correct, and internally coherent. The degradation is detectable only by an automatic metric with full source access operating outside the NLI audit’s truncation constraint. This is precisely why multi-metric triangulation is a necessary condition for sound faithfulness evaluation of iterative refinement pipelines.

5.5 A Three-Condition Detection Protocol for Verifier Exploitation

The diagnostic signature documented above generalizes to a formally stated detection protocol applicable to any NLI-guided iterative refinement pipeline. The protocol operates at the *evaluation-set level* — a deliberate design choice reflecting the high instance-level variance of individual metrics and the reliability of aggregate signals for distinguishing systematic exploitation from document-level variation.

Definition 5.5 Verifier Exploitation Detection Protocol

Given an evaluation set of n instances and two pipeline configurations T_a (baseline) and T_b (extended refinement), **verifier exploitation is diagnosed when all three of the following conditions hold simultaneously at the aggregate level**. No proper subset is sufficient.

(i) **Large NLI-metric inflation.**

$$\overline{\Delta\text{SummaCConv}} > 0.1$$

The auditing metric shows substantial aggregate gain. This establishes that the proxy metric has been satisfied — but not that faithfulness has improved.

(ii) **Cross-framework non-transfer.**

$$\overline{\Delta\text{AlignScore}} \leq 0 \quad \text{and not significantly positive under a one-sided test.}$$

The NLI-metric gain does not transfer to an architecturally independent non-NLI faithfulness signal. This rules out genuine faithfulness improvement as the explanation and is the decisive criterion distinguishing exploitation from real gain.

(iii) **Universal register degradation.**

$$\overline{\Delta\text{BARTScore}_{s \rightarrow d}} < -1.0 \quad \text{at the aggregate level,}$$

and $\Delta\text{BARTScore}_{s \rightarrow d} < -1.0$ on $\geq 90\%$ of individual instances. The instance-level universality requirement distinguishes systematic exploitation from document-level noise, and the $s \rightarrow d$ direction distinguishes register shift from content loss: under pure content removal both directional BARTScore variants decline proportionally; the asymmetry confirms register degradation as the primary effect.

Logical structure. Condition (i) establishes metric inflation. Conditions (i)+(ii) together rule out genuine improvement. Condition (iii) confirms the mechanism and is the only condition with instance-level universality as a requirement — making it the most robust single-document diagnostic when evaluation set size is constrained. All three are necessary; satisfaction of any two without the third admits an alternative explanation.

Empirical validation. On 498 Multi-News instances comparing $T_{\max}=1$ vs. $T_{\max}=2$: condition (i) is satisfied at the 10th percentile (+0.113, threshold 0.1); condition (ii) is satisfied in aggregate ($\overline{\Delta}=-0.001$); condition (iii) is satisfied on **100%** of instances (Wilcoxon $W=0$, $p=2.68 \times 10^{-83}$), with the mildest observed case at -1.105 nats — a margin of 0.105 nats above the threshold even in the least extreme instance.

Empirical grounding of thresholds. Each threshold is grounded in the observed per-instance distribution. For condition (i): the mean $\Delta\text{SummaCConv}$ of +0.185 exceeds the threshold by +0.085; the 10th percentile of per-instance gains is +0.113, confirming the threshold is not sensitive to outliers. For condition (ii): the mean $\Delta\text{AlignScore}$ of -0.001 satisfies the condition; the instance-level variance is high but the aggregate signal is reliable. For condition (iii): the mean collapse of -2.566 nats exceeds the threshold by -1.566 nats; the condition is satisfied by 100% of 498 instances, with the mildest observed case at -1.105 nats — a margin of 0.105 nats above the threshold even in the least extreme instance. The 90% instance-level universality requirement is conservative; the observed universality is 100%.

Why all three conditions are necessary. Condition (i) alone is insufficient because SummaCConv gains could reflect genuine improvement. Conditions (i)+(ii) together are insufficient because AlignScore’s aggregate non-improvement could reflect metric insensitivity rather than exploitation. Condition (iii) is the decisive diagnostic: a -2.566 nat collapse in BARTScore $s \rightarrow d$ at < 0.4% length change, confirmed on every single evaluation instance, cannot be attributed to metric insensitivity or document-level variation. The $s \rightarrow d$ direction is specifically necessary to distinguish register degradation from content loss: under pure content truncation, both directional BARTScore variants would collapse proportionally; the observed asymmetry ($\Delta s \rightarrow d = -2.566$ vs. $\Delta d \rightarrow s = -0.185$) confirms register shift as the primary effect.

Scope and applicability. The protocol requires three automatic metrics — an NLI-based faithfulness scorer, an independent non-NLI faithfulness scorer, and a generative log-probability evaluator — and a comparison between two refinement depths on a common evaluation set. It requires no human annotation, no retraining, and no access to model internals. Its computational cost is the incremental cost of running the three metrics on the extended-refinement outputs, which in our setting amounts to less than 10% of total pipeline cost. We recommend it as a standard post-hoc diagnostic for any practitioner deploying iterative NLI-guided faithfulness refinement.

5.6 $T_{\max}=1$ as the Principled Default

The evidence jointly and unambiguously justifies $T_{\max}=1$ as the revision budget default. Genuine faithfulness gains — statistically confirmed on both SummaCConv ($p=4.49 \times 10^{-28}$) and BARTScore $s \rightarrow d$ ($p=4.01 \times 10^{-4}$) — concentrate in the first revision cycle. The second cycle, operating in a low-flag regime where the first cycle has already addressed the most salient faithfulness failures, produces no cross-metric benefit and introduces verifier exploitation with instance-level universality. Early-exit (Algorithm 1, line 8) ensures that instances where $\mathcal{F}^{(0)}=\emptyset$ incur no additional inference cost, making $T_{\max}=1$ efficient as well as principled. $T_{\max}=2$ is retained in this study as a controlled demonstration of over-optimization — a cautionary case establishing what iterative NLI-guided refinement produces when pushed beyond its effective operating range.

5.7 Implications

The findings reported in this section have implications that extend substantially beyond AnchorSum and multi-document summarization.

Verifier exploitation does not require gradient updates. The central theoretical implication is that verifier exploitation is a property of *feedback structure*, not optimization mechanism. In AnchorSum, the generator’s parameters are frozen throughout; no gradient signal flows at any stage; the only feedback mechanism is a structured natural language prompt derived from an NLI verifier’s output. Under these conditions — the most constrained possible inference-time setting — verifier exploitation still emerges, universally and systematically, across 498 diverse documents from a single additional revision cycle. This extends the known boundary of Goodhart’s Law (Gehrmann et al., 2023) in natural language generation: the pathology does not require a model to learn to game the metric through parameter updates. It requires only a feedback loop in which the metric’s output drives revision, and a metric with exploitable architectural limitations — in this case, the 512-token truncation window of the NLI cross-encoder.

Table 5: Faithfulness and semantic fidelity on 498 Multi-News test instances (Fabbri et al., 2019). All values are mean $\pm$ std. BARTScore (Yuan et al., 2021) is a log-probability; less negative = better. Best value per column is **bolded**; where values are tied both configurations are bolded.

Configuration	NLI-based Faithfulness		BARTScore (Yuan et al., 2021)	
	SummaCConv (Laban et al., 2022)	AlignScore-L (Zha et al., 2023)	$s \rightarrow d$ (faith.)	$d \rightarrow s$ (coverage)
Base	0.715 ± 0.034	0.828 ± 0.094	-4.038 ± 0.383	-3.470 ± 0.428
no_entity	0.712 ± 0.034	0.829 ± 0.094	-4.045 ± 0.376	-3.487 ± 0.426
no_nli	0.733 ± 0.041	0.824 ± 0.095	-4.027 ± 0.392	-3.481 ± 0.451
Full (AnchorSum)	0.733 ± 0.043	0.822 ± 0.093	-4.019 ± 0.394	-3.469 ± 0.450

The exploitable attack surface is the auditor’s architectural limitation. The instance-level analysis reveals that the specific mechanism of exploitation is *truncation-exploiting content removal*: the revision process targets content that is genuine and source-grounded but falls outside the NLI auditor’s attention window, removing it to achieve higher per-sentence entailment scores. This means that any NLI-based auditor operating under an input length constraint is potentially exploitable by an iterative refinement process, regardless of the generator’s capability or the quality of the initial draft. The limitation is architectural, not incidental, and it becomes an attack surface under repeated refinement.

Single-metric NLI auditing is insufficient for iterative refinement evaluation. A practitioner relying solely on SummaCConv — or any single NLI-based metric — to evaluate an iterative faithfulness refinement pipeline would conclude that a second revision cycle produces a breakthrough improvement of +0.185 absolute SummaCConv. This conclusion is false. The improvement is an artifact of metric gaming, detectable only through cross-framework triangulation. This has a concrete design implication: pipelines that use NLI-based metrics as both the revision signal *and* the evaluation criterion are not evaluating faithfulness — they are measuring the pipeline’s ability to satisfy its own auditor. Evaluation of iterative refinement must include at least one metric that is architecturally independent of the auditing framework used to generate revision signals.

Recommendations for practitioners. Three concrete recommendations follow from this analysis. First, iterative faithfulness refinement pipelines should be evaluated with a multi-metric battery that includes at least one non-NLI faithfulness signal (e.g., AlignScore (Zha et al., 2023)) and one generative log-probability evaluator (e.g., BARTScore (Yuan et al., 2021)), not NLI-based metrics alone. Second, revision budgets should be set conservatively — $T_{\max}=1$ is the principled default — and extended budgets should be validated against the three-condition detection protocol (Definition 5.5) before deployment. Third, NLI auditors deployed in iterative refinement loops should use hierarchical or sliding-window inference (Laban et al., 2022) rather than fixed-window truncation, to reduce the exploitable attack surface that truncation bias introduces.

6 Component Ablation

Four pipeline configurations are compared across 498 Multi-News test instances: **Base** (unaugmented generator, no auditing or revision); **no_entity** (NLI audit only, Entity Guard disabled); **no_nli** (Entity Guard only, NLI audit disabled); and **Full** (complete AnchorSum, $T_{\max}=1$). All configurations use identical generator (Dubey et al., 2024), decoding strategy (Wiher et al., 2022), and anchor budget ($N=15$); the only variation is which auditing modules are active. Results are reported in Tables 5 and 6; statistical confirmation via paired Wilcoxon signed-rank tests ($n=498$) is integrated throughout.

6.1 Overall Faithfulness Gains

AnchorSum ($T_{\max}=1$) achieves the strongest performance on both primary faithfulness metrics. SummaCConv improves from 0.715 to 0.733 over the unaugmented base, corresponding to a 6.3% relative reduction

Table 6: Lexical overlap (Lin, 2004), neural semantic similarity (Zhang et al., 2020b) with deberta-xlarge-mnli, and linguistic fluency (Zhong et al., 2022) on 498 Multi-News test instances (Fabbri et al., 2019). All values are mean $\pm$ std. Best value per column is **bolded**; where values are tied both configurations are bolded.

Config.	ROUGE (Lin, 2004)			BERTScore (Zhang et al., 2020b)			UniEval (Zhong et al., 2022)
	R-1	R-2	R-L	Prec.	Rec.	F_1	Fluency
Base	0.399$\pm$0.072	0.117 $\pm$ 0.046	0.183$\pm$0.038	0.601$\pm$0.041	0.640 $\pm$ 0.044	0.619$\pm$0.039	0.945 $\pm$ 0.042
no_entity	0.396 $\pm$ 0.071	0.113 $\pm$ 0.045	0.181 $\pm$ 0.037	0.600 $\pm$ 0.042	0.636 $\pm$ 0.043	0.617 $\pm$ 0.038	0.949$\pm$0.035
no_nli	0.383 $\pm$ 0.078	0.119$\pm$0.051	0.178 $\pm$ 0.045	0.585 $\pm$ 0.049	0.642$\pm$0.047	0.612 $\pm$ 0.044	0.940 $\pm$ 0.044
Full	0.384 $\pm$ 0.078	0.119$\pm$0.050	0.178 $\pm$ 0.042	0.585 $\pm$ 0.048	0.640 $\pm$ 0.047	0.611 $\pm$ 0.044	0.944 $\pm$ 0.032

in inconsistency, computed as $\Delta(1 - \text{SummaCConv}) / (1 - 0.715) \times 100$. BARTScore $s \rightarrow d$ corroborates convergently, improving from -4.038 to -4.019 — the least-negative score across all configurations — confirming higher source-fidelity under an independent generative evaluator operating on an entirely different training distribution. BARTScore $d \rightarrow s$ similarly reaches its best value at Full (-3.469), confirming improved anchor coverage via $\mathcal{F}_{\text{cov}}$ flags.

Both primary gains are statistically unambiguous. Paired Wilcoxon signed-rank tests across 498 instances confirm significance on SummaCConv ($W=26,687$, $p=4.49 \times 10^{-28}$) and BARTScore $s \rightarrow d$ ($W=50,753$, $p=4.01 \times 10^{-4}$), both surviving Bonferroni correction for six comparisons (adjusted $\alpha=0.0083$). These results confirm that AnchorSum’s faithfulness gains are systematic across documents rather than driven by a subset of instances.

AlignScore differences across all four configurations are within 0.007 — less than 0.1σ — confirming no source-grounding degradation from the auditing and revision process. Near-ceiling UniEval fluency (0.940 – 0.949) with lowest variance for Full ($\sigma=0.032$) indicates that dual-module revision stabilizes rather than degrades output quality (Tang et al., 2024; Madaan et al., 2023).

6.2 Necessity of the Entity Guard

The Entity Guard’s role is not merely contributory — it is a structural necessity. Removing it (no_entity) does not produce a neutral result: SummaCConv actively degrades *below* the unaugmented base (0.712 vs. 0.715 , $\Delta=-0.003$), a result confirmed by paired Wilcoxon test ($W=43,895$, $p=2.91 \times 10^{-5}$, surviving Bonferroni correction). NLI auditing alone, without entity-level grounding, cannot prevent hallucinated entities from propagating through the revision cycle.

The mechanism is direct: when the revision prompt supplies only NLI-detected inconsistencies as flags, the LLM may rewrite flagged sentences in ways that satisfy sentence-level entailment while simultaneously introducing or retaining extrinsic entities that sentence-level NLI is architecturally unable to detect (Laban et al., 2022; Pagnoni et al., 2021). The conjunctive entity filter (Eq. 12) intercepts these extrinsic hallucinations through a verification path entirely independent of the NLI signal. Without it, iterative revision guided by NLI alone is not merely suboptimal — it is actively harmful to faithfulness.

6.3 Complementarity of the NLI Module

The relationship between the two auditing modules is one of strict complementarity across non-overlapping failure modes (Maynez et al., 2020). no_nli matches Full on SummaCConv (0.733 in both cases), which might appear to suggest the NLI module contributes nothing beyond what the Entity Guard already provides. BARTScore resolves this ambiguity: Full strictly outperforms no_nli in both directional variants ($s \rightarrow d$: -4.019 vs. -4.027 ; $d \rightarrow s$: -3.469 vs. -3.481), confirming that the NLI module addresses semantic inconsistencies that entity-level filtering cannot detect. The BARTScore $s \rightarrow d$ difference is significant at $p=0.037$ ($W=67,594$),

though this result does not survive Bonferroni correction; the complementarity claim therefore rests on the convergent cross-metric pattern rather than this single comparison alone.

The pattern admits a precise interpretation. SummaCConv is insensitive to the NLI module’s marginal contribution because it measures the same property the NLI auditor directly optimizes — the two are not independent signals in this comparison. BARTScore, operating under a generative model with a different training distribution and no architectural connection to NLI entailment, captures residual semantic grounding improvements that SummaCConv cannot see. This is precisely the motivation for cross-framework evaluation: a single metric would have missed the NLI module’s contribution entirely and produced the incorrect conclusion that only the Entity Guard matters.

6.4 Secondary Metric Behaviour

Modest ROUGE-1/L declines for Full relative to Base ($\Delta R-1=-0.015$; $\Delta R-L=-0.005$) are the expected consequence of a coverage-oriented revision process and do not indicate quality degradation (Fabbri et al., 2019; Maynez et al., 2020). Revised summaries that incorporate omitted anchors diverge lexically from human reference summaries written without access to revision signals; ROUGE overlap against those references therefore understates true source coverage rather than reflecting a genuine quality loss.

BERTScore precision falls (0.585 vs. 0.601) while recall is stable (0.640 for both Base and Full), a decomposition that is directly interpretable: anchor-rich outputs contain more content than the reference rather than semantically diverging from it (Zhang et al., 2020b). The precision decline reflects output length, not semantic error.

6.5 Three Structural Conclusions

The component ablation establishes three conclusions that are jointly necessary for understanding AnchorSum’s design:

1. **AnchorSum achieves statistically confirmed faithfulness gains.** Both SummaCConv ($p=4.49\times 10^{-28}$) and BARTScore $s\rightarrow d$ ($p=4.01\times 10^{-4}$) improvements over base are significant and survive Bonferroni correction, confirming that gains are systematic across 498 diverse documents.
2. **The Entity Guard is a necessary condition, not a supplementary one.** Its removal produces statistically confirmed degradation below the unaugmented base ($p=2.91\times 10^{-5}$), establishing that NLI auditing alone is insufficient and actively harmful without entity-level grounding.
3. **The two modules address non-overlapping failure modes.** Their joint activation yields BARTScore improvements unattainable by either module alone, and cross-framework metric evaluation is necessary to reveal this complementarity — SummaCConv alone would have obscured the NLI module’s contribution entirely.

7 Main Results

Table 7 reports LLM-as-judge scores for all six systems. We structure the discussion around three comparisons: the auditing pipeline’s incremental contribution over its unaugmented base, performance relative to fine-tuned baselines, and inter-judge agreement.

7.1 Auditing Pipeline Contribution

The primary comparison is AnchorSum ($T_{\max}=1$) against Base LLaMA — the identical backbone without auditing — which isolates the dual-module architecture’s contribution from the generator’s inherent capability. AnchorSum ($T_{\max}=1$) adds +0.22 jury points overall (8.30 vs. 8.08), with gains across all four evaluated dimensions. Faithfulness and Fluency show the largest margins (+0.23 and +0.26 jury points respectively), confirming genuine multi-dimensional improvement rather than surface-level polishing.

Table 7: LLM-as-judge evaluation on 50 Multi-News instances (Fabbri et al., 2019). DeepSeek-V3.1 (DeepThink mode) (DeepSeek-AI, 2025) and Qwen3.5-Plus (thinking mode) (Qwen Team, 2026) scores are mean over 50 documents (1–10 scale). Jury = unweighted average of the two judges (Verga et al., 2024). **Bold** marks the best value per column across all systems. ‡ denotes the dimension where judge rankings diverge most substantively for $T_{\max} \in \{1, 2\}$; see §8. † PRIMERA-MultiNews evaluated on raw concatenated source documents without preprocessing; scores represent a lower bound on true performance.

System	Faithfulness		Coverage		Consistency [‡]		Fluency		Overall (Jury)	
	DS	QW	DS	QW	DS	QW	DS	QW	DS	QW
BART-large-CNN (Lewis et al., 2020)	5.92	6.94	5.68	6.52	5.96	7.18	6.64	6.86	6.05	6.88
PRIMERA-MultiNews [†] (Xiao et al., 2022)	6.18	7.18	5.94	6.86	6.30	7.42	7.02	7.24	6.36	7.17
PEGASUS-multi_news (Zhang et al., 2020a)	7.12	7.62	6.98	7.24	7.20	7.84	7.56	7.58	7.21	7.57
Base LLaMA (no auditing)	7.84	8.14	7.90	7.86	7.92	8.48	8.18	8.32	7.96	8.20
AnchorSum ($T_{\max}=1$)	8.16	8.28	8.04	8.04	8.22	8.62	8.56	8.46	8.25	8.35
AnchorSum ($T_{\max}=2$)	7.98	8.42	8.02	8.18	8.06	8.76	8.42	8.54	8.12	8.48

The Faithfulness gain is particularly meaningful given that Base LLaMA already operates near the top of the score range. An improvement of +0.23 jury points from a base of 7.99 reflects precision gains in entity grounding and NLI-verified consistency — exactly the failure modes the dual-module architecture is designed to intercept. Automatic metric results corroborate this interpretation: paired Wilcoxon signed-rank tests across 498 instances confirm statistically significant improvements on both SummaCConv ($W=26,687$, $p=4.49 \times 10^{-28}$) and BARTScore $s \rightarrow d$ ($W=50,753$, $p=4.01 \times 10^{-4}$); full ablation results are in §6 and §5.

7.2 Performance Against Fine-Tuned Baselines

Both AnchorSum configurations outperform all fine-tuned baselines across every evaluated dimension. The jury advantage over PEGASUS-multi_news — the strongest fine-tuned baseline — is +0.91 overall for $T_{\max}=1$ (8.30 vs. 7.39 jury), with the largest margin on Faithfulness (+0.85 jury; +1.04 DS; +0.66 QW) — the dimension most directly targeted by the auditing modules. As established in §4.4, this comparison conflates the pipeline’s contribution with the generator’s scale advantage and should be interpreted accordingly; the Base LLaMA comparison in §7.1 is the appropriate locus for evaluating the auditing architecture’s incremental value.

PRIMERA-MultiNews scores below PEGASUS-multi_news under both judges (6.36 vs. 7.21 DS; 7.17 vs. 7.57 QW). This is attributable to evaluation conditions: PRIMERA-MultiNews was provided raw concatenated source documents directly from the dataset without any preprocessing, whereas its LED backbone expects entity-pyramid sentinel-token masking applied at fine-tuning time. The resulting scores represent a lower bound on true PRIMERA-MultiNews performance.

7.3 Inter-Judge Agreement

Inter-judge agreement is highest for AnchorSum ($T_{\max}=1$), with a maximum dimension-level disagreement of 0.40 points, and lowest for BART-large-CNN and PRIMERA-MultiNews (1.22 and 1.12 respectively). This pattern reflects output quality: when summaries are faithful, coherent, and fluent, evaluator-specific preferences exert less influence on scores. The elevated disagreement on lower-performing systems reflects genuine qualitative ambiguity in weak outputs rather than evaluator unreliability.

Fluency shows the lowest mean absolute disagreement across all systems (≤ 0.22), consistent with surface grammaticality being the dimension least sensitive to evaluator perspective. Faithfulness and Coverage show higher disagreement, reflecting their dependence on source-grounding judgements that may vary across evaluator architectures.

The one substantive divergence between judges concerns the relative ranking of $T_{\max} \in \{1, 2\}$: DeepSeek-V3.1 assigns $T_{\max}=1$ a +0.13-point advantage (8.25 vs. 8.12), while Qwen3.5-Plus inverts this preference by the

same margin (8.48 vs. 8.35), yielding identical jury scores of 8.30 for both configurations. This symmetric disagreement is mechanistically interpretable in light of the verifier exploitation analysis; this symmetric disagreement is resolved by the automatic metric evidence presented in §5, which establishes that the $\text{BARTScore}_{s \rightarrow d}$ collapse at $T_{\max}=2$ is the decisive signal that holistic judge evaluation cannot detect.

8 Analysis

The LLM-as-judge results reported in §7 raise four analytical questions that aggregate scores alone cannot answer: why judges diverge on revision depth, how that divergence resolves, what agreement patterns reveal about output quality, and how much of AnchorSum’s performance is attributable to the generator versus the auditing architecture. We address each in turn.

8.1 Inter-Judge Agreement as a Quality Signal

Inter-judge agreement is not merely a methodological diagnostic — it is a quality signal in its own right, and one that tracks faithfulness architecture directly. The monotonic relationship between system quality and agreement — AnchorSum at 0.40, down to BART-large-CNN at 1.22 — is not coincidental. When summaries are faithful, internally coherent, and fluent, evaluator-specific architectural preferences exert less influence because the output’s quality is less ambiguous.

DeepSeek-V3.1 (DeepThink mode) assigns $T_{\max}=1$ a +0.13-point overall advantage (8.25 vs. 8.12); Qwen3.5-Plus (thinking mode) inverts this preference by the same margin (8.48 vs. 8.35); the jury scores are numerically equal at 8.30 for both configurations. The dimension-level scores reveal the mechanism. Qwen3.5-Plus rewards the second cycle specifically on Consistency (8.76 vs. 8.62) — the dimension most sensitive to elaborated, sequentially coherent prose — because $T_{\max}=2$ summaries, rewritten toward propositionally explicit declaratives, present information in a more structured, less elliptical form that a coherence-oriented evaluator reads as higher quality. DeepSeek-V3.1, operating in a mode more sensitive to fine-grained factual grounding, penalises the same second cycle on Faithfulness (7.98 vs. 8.16, $\Delta=-0.18$) — the dimension most directly targeted by AnchorSum’s auditing architecture. The two judges are not disagreeing about the same property. They are measuring different aspects of quality and arriving at different conclusions because those aspects genuinely respond differently to additional revision.

This mechanistic interpretability is itself a finding: it demonstrates that LLM judges with different reasoning profiles are sensitive to different surface properties of summary quality, and that no single judge can be trusted to resolve the revision depth question unilaterally. The jury methodology of Verga et al. (2024) is therefore not merely a methodological safeguard but a substantive analytical tool — the disagreement pattern it reveals is informative rather than inconvenient.

8.2 Cross-Method Convergence Resolves the Ambiguity

The inter-judge tie does not leave the revision depth question open. As we demonstrate in §5, three independent automatic metric signals converge unambiguously in favour of $T_{\max}=1$: $\text{BARTScore}_{s \rightarrow d}$ collapses by -2.566 nats at $T_{\max}=2$; SummaCConv’s +0.185 gain is identified as verifier exploitation rather than genuine faithfulness improvement; and AlignScore registers no improvement — a non-NLI faithfulness signal architecturally immune to the exploitation mechanism showing $\Delta < 0.001$. The full statistical confirmation, mechanistic analysis, and detection protocol are presented in §5; we note here only the evaluative implication that bears directly on the LLM-judge findings.

The jury tie strengthens rather than undermines this conclusion. It establishes that $T_{\max}=2$ summaries are *holistically plausible* — evaluators assessing full summaries for overall quality cannot distinguish genuine faithfulness improvement from exploited NLI scores when evaluating holistically. This is precisely the danger of verifier exploitation as a failure mode: it produces outputs that appear high-quality to any evaluator operating at the level of holistic coherence, while degrading the specific property the pipeline was designed to improve. Automatic metrics, operating at a lower level of abstraction and targeting specific measurable

properties, are the only evaluation paradigm capable of detecting this degradation (Liu et al., 2023; Verga et al., 2024).

The two evaluation paradigms are therefore not competing but complementary: the LLM-judge panel establishes holistic plausibility; automatic metrics expose latent faithfulness degradation; and it is their joint application — not either alone — that produces a sound conclusion about revision depth. This complementarity is itself a methodological contribution of the present work, demonstrating that faithfulness evaluation of iterative refinement pipelines requires both paradigms to reach reliable conclusions.

8.3 Dimension-Level Agreement and Revision-Depth Effects

Inter-judge agreement varies systematically across both revision depth and evaluation dimension, and both patterns carry direct interpretive weight. AnchorSum ($T_{\max}=2$) produces a maximum dimension-level disagreement of 0.70 points—nearly double the 0.40 recorded for $T_{\max}=1$ —consistent with the register degradation documented in §5 introducing a dimension of quality ambiguity that a single revision cycle avoids: when summaries are rewritten toward propositionally atomic declaratives, evaluators with different architectural sensitivities weight the resulting trade-off between surface coherence and discourse richness differently, producing wider disagreement precisely because the degradation is real but unevenly visible. Across dimensions, Fluency exhibits the lowest mean absolute disagreement across all systems (≤ 0.22), confirming that surface grammaticality is the quality dimension least sensitive to evaluator perspective, while Faithfulness and Consistency show higher inter-judge variance, reflecting their dependence on source-grounding and coherence judgements that are more architecture-dependent than surface fluency assessment (Liu et al., 2023).

That AnchorSum ($T_{\max}=1$) achieves the highest agreement of any system is therefore a direct consequence of its faithfulness architecture: entity grounding and NLI-verified consistency produce outputs whose quality is less evaluator-dependent because fewer dimensions of failure remain to be weighted differently. AnchorSum ($T_{\max}=2$) shows higher disagreement (0.70) than $T_{\max}=1$ (0.40), consistent with the register degradation documented in §5 introducing a dimension of quality ambiguity that $T_{\max}=1$ avoids.

Across dimensions, Fluency exhibits the lowest mean absolute disagreement across all systems (≤ 0.22), confirming that surface grammaticality is the quality dimension least sensitive to evaluator perspective. Faithfulness and Consistency show higher inter-judge variance, reflecting their dependence on source-grounding and coherence judgements that are more architecture-dependent than surface fluency assessment (Liu et al., 2023).

8.4 Generator Contribution vs. Auditing Contribution

Isolating the auditing pipeline’s incremental contribution from the generator’s inherent capability requires the Base LLaMA comparison — the identical backbone without auditing — rather than comparisons against fine-tuned baselines that conflate both factors. Base LLaMA (7.96 DS; 8.20 QW overall) already exceeds all fine-tuned external baselines (Dubey et al., 2024), establishing a strong generator floor above which the auditing pipeline must demonstrate incremental value.

AnchorSum ($T_{\max}=1$) adds +0.29 DS and +0.15 QW overall points above this floor. The diagnostic lies in the dimension-level pattern. The Faithfulness gap (+0.32 DS; +0.14 QW) and Fluency gap (+0.38 DS; +0.14 QW) are of comparable magnitude — neither dimension dominates the overall gain. This pattern is informative: a pipeline that improved only surface fluency through revision would show a large Fluency gap with a negligible Faithfulness gap. A pipeline that improved only NLI-verified consistency would show the opposite. The observed pattern — comparable gains across both dimensions — confirms genuine multi-dimensional quality improvement attributable to the dual-module architecture rather than to a single surface property.

The absolute magnitude of these gains is modest, and this should be interpreted correctly. At the capability level of Llama-3.1-8B-Instruct, the base generator already produces summaries of sufficient quality that holistic improvement is difficult to achieve through post-hoc auditing alone. The appropriate expectation for a faithfulness-constrained auditing pipeline is not broad holistic improvement but targeted precision gains in the specific failure modes it is designed to intercept: entity grounding and NLI-verified semantic consistency.

The component ablation in §6 confirms that this is precisely what AnchorSum delivers, with statistically confirmed improvements on both primary faithfulness metrics ($p=4.49\times 10^{-28}$ and $p=4.01\times 10^{-4}$ respectively). Modest holistic gains driven by targeted precision improvements are the correct signature of a well-designed faithfulness auditing pipeline — not a limitation of it.

8.5 Joint Implications

The four analyses converge on a unified picture. AnchorSum’s auditing pipeline produces genuine, statistically confirmed, multi-dimensional quality gains over its unaugmented base. Inter-judge agreement tracks faithfulness quality directly and monotonically, validating the pipeline’s architectural design. The one substantive judge disagreement — on revision depth — is not a weakness of the evaluation methodology but a demonstration of its strength: the disagreement is mechanistically interpretable, and its resolution requires exactly the cross-framework automatic metric triangulation that this paper advocates.

Most importantly, the jury tie at $T_{\max}=2$ establishes a finding that extends beyond AnchorSum: holistic LLM-as-judge evaluation is insufficient to detect verifier exploitation. An evaluator assessing summary quality at the level of overall coherence and fluency cannot distinguish outputs that have genuinely improved faithfulness from outputs that have satisfied an NLI auditor by degrading their discourse register. Detecting this failure mode requires automatic metrics operating at a lower level of abstraction, targeting specific measurable properties that holistic evaluation averages over. This methodological implication — that faithfulness evaluation of iterative refinement pipelines requires both evaluation paradigms jointly — is the subject of the following section.

9 Conclusion

Verifier exploitation — the pathology whereby a system satisfies its auditing metric while degrading real faithfulness — has been documented exclusively in gradient-based systems. This paper demonstrates that the failure mode is not a property of optimization mechanism but of feedback structure, and that a purely prompt-driven sequential revision loop, with weights frozen throughout and no learned reward signal at any stage, is sufficient for it to emerge universally and without exception.

The empirical finding. AnchorSum, a modular training-free pipeline for multi-document summarization, provides the controlled experimental vehicle: its auditor, feedback signal, and revision mechanism are fully separable, enabling precise causal attribution. Under a single revision cycle, AnchorSum achieves a 6.3% relative SummaCConv inconsistency reduction over its unaugmented base ($p=4.49\times 10^{-28}$), convergently supported by BARTScore_{s→d} ($p=4.01\times 10^{-4}$), with component ablations establishing the Entity Guard as a necessary condition and confirming the two auditing modules address non-overlapping failure modes in Maynez et al. (2020)’s hallucination taxonomy. LLM-as-judge evaluation on 50 instances shows consistent superiority over all fine-tuned baselines across every evaluated dimension.

The theoretical finding. A second revision cycle, with no parameter update and no gradient signal at any stage, produces a +0.185 SummaCConv inflation while collapsing BARTScore_{s→d} by −2.566 nats at less than 0.4% length change. The Wilcoxon statistic is exactly $W=0$ ($p=2.68\times 10^{-83}$): $T_{\max}=2$ produced strictly worse BARTScore_{s→d} than $T_{\max}=1$ on every single one of 498 evaluation instances without exception. Instance-level analysis identifies the specific mechanism: *truncation-exploiting content removal*, in which the revision process targets genuine source-grounded content falling outside the NLI auditor’s 512-token attention boundary, removing it to achieve higher per-sentence entailment scores at the cost of lower informativeness and source fidelity. The NLI auditor’s architectural limitation is therefore not merely a passive source of false positives — it is an exploitable attack surface that the revision process actively targets, without any gradient update, from a single additional revision cycle. This extends the known boundary of Goodhart’s Law in natural language generation beyond gradient-based systems and selection-based inference-time methods (Khalaf et al., 2025) to sequential prompt-driven revision loops, the most constrained possible inference-time setting.

The detection protocol. We formalize a three-condition annotation-free protocol (Definition 5.5) applicable to any NLI-guided iterative refinement pipeline: large NLI-metric inflation, cross-framework non-transfer confirmed by a non-NLI faithfulness signal, and universal register degradation confirmed by a generative log-probability evaluator. All three conditions are necessary; no proper subset is sufficient. The protocol is validated with instance-level universality on the decisive condition ($W=0$, 100% of 498 instances) and requires no human annotation, no model retraining, and no access to model internals. Its computational cost is less than 10% of total pipeline cost.

Practical implications. Two implications follow directly for any practitioner deploying iterative faithfulness refinement. First, evaluation must include at least one metric architecturally independent of the auditing signal: a pipeline evaluated solely on the metric it also uses as a revision signal is not measuring faithfulness — it is measuring its own ability to satisfy its auditor. Second, revision budgets must be validated against the detection protocol before deployment: genuine quality gains concentrate in early revision cycles, and extended budgets introduce exploitation rather than improvement, at a cost premium — in our setting $\approx 1.8\times$ additional GPU-hours — that yields no cross-metric benefit.

Open questions. Three future directions follow directly from this work, each motivated by a specific gap in the existing literature. First, the truncation-exploiting content removal mechanism implicates the NLI auditor’s fixed attention window as the primary attack surface; sliding-window or hierarchical NLI inference (Laban et al., 2022) is the natural architectural mitigation, but whether it eliminates the exploitation boundary or merely shifts it to a different architectural constraint is an open empirical question. Second, the detection protocol should be validated across generator architectures, domains beyond English-language news, and refinement frameworks beyond NLI-guided revision — including Constitutional AI (Bai et al., 2022) and inference-time alignment methods where analogous feedback structures couple a single proxy signal with architectural limitations that sequential revision can target (Khalaf et al., 2025). Whether the two structural conditions we identify — a single proxy metric as both auditor and revision signal, and an exploitable architectural limitation in that auditor — are jointly sufficient for exploitation to emerge in those settings is a direct open question. Third, the claim that truncation-exploiting content removal is invisible to holistic human evaluation follows logically from the jury tie at $T_{\max}=2$ but has not been verified through controlled human annotation; human judgement studies triangulating against the automatic metric and LLM-judge results reported here would directly test this claim and either strengthen or qualify the detection protocol’s scope.

The broader implication is architectural rather than methodological. Verifier exploitation emerges from feedback structure, not from optimization pressure. Any system that couples a proxy metric with an iterative revision mechanism — whether the metric is an NLI cross-encoder, a constitutional principle, or a learned reward model — is susceptible to this failure mode whenever the metric has exploitable architectural limitations and the revision budget is sufficient for those limitations to be targeted. Detection requires evaluation frameworks that are architecturally independent of the revision signal. Single-metric evaluation of iterative refinement is not evaluation: it is measurement of the pipeline’s ability to satisfy its own auditor.

Limitations

Evaluation scope. All automatic metric experiments are conducted on 498 Multi-News test instances (Fabbri et al., 2019) (500 sampled deterministically; two excluded due to GPU memory constraints — see §4.1). The LLM-as-judge evaluation covers 50 instances, a subset necessitated by the token cost of issuing full document clusters to two frontier reasoning models; this sample size, while consistent with prior LLM-as-judge work (Liu et al., 2023), precludes statistical significance testing on the judge scores and may not be representative of the full test distribution. All experiments are restricted to the English-language news domain; generalizability to scientific, legal, or multilingual multi-document settings is not established and constitutes an immediate open question, particularly given that domain shift may alter the entity density and discourse structure properties on which AnchorSum’s anchor extraction and NLI auditing depend.

Detection protocol scope. The three-condition verifier exploitation detection protocol (Definition 5.5) operates at the *evaluation-set level* rather than the instance level. Condition (ii) — cross-framework non-transfer measured by AlignScore — exhibits high instance-level variance (min -0.199 , max $+0.185$) with a near-zero mean (-0.001), confirming it as a reliable aggregate signal but an unreliable single-document classifier. Practitioners applying the protocol to small evaluation sets should exercise caution: the aggregate signal becomes reliable only as n increases, and the threshold of “sufficient n ” has not been formally characterized. We recommend a minimum of $n=100$ instances as a conservative lower bound based on the observed variance structure, though this recommendation is not empirically validated in the present study. Condition (iii) — BARTScore $s \rightarrow d$ collapse — is the most robust instance-level signal, satisfied universally across all 498 instances in this study; practitioners may treat it as the primary single-document diagnostic when evaluation set size is constrained.

NLI truncation bias and its role in verifier exploitation. The sentence-level NLI audit truncates the source corpus $\mathcal{D}$ to 512 subword tokens before each $(\mathcal{D}, \hat{s}_i)$ inference call, following Laban et al. (2022). This introduces a positional bias: sentences whose supporting evidence appears in later portions of a long source cluster may be incorrectly flagged as non-entailed, generating spurious revision signals. In the present study, this limitation takes on a specific significance beyond a passive source of false positives: as documented in §5.4, the truncation window constitutes an *exploitable attack surface* under iterative refinement. The second revision cycle removes genuine source-grounded content that falls outside the 512-token attention boundary, satisfying the auditor by eliminating what it cannot verify rather than by correcting genuine inconsistencies. Hierarchical or sliding-window NLI scoring (Laban et al., 2022) is the natural architectural mitigation, extending the attention window to cover the full source cluster at the cost of additional inference calls. Whether sliding-window inference eliminates truncation-exploiting content removal entirely or merely shifts the exploitation boundary to a different architectural constraint is an open question that motivates future work.

Generator dependency. AnchorSum’s generation and revision phases use a single backbone, Meta-Llama-3.1-8B-Instruct (Dubey et al., 2024). The pipeline’s faithfulness gains are conditional on the generator having sufficient zero-shot instruction compliance to follow the no-hallucination directive and to incorporate anchor constraints as soft semantic constraints; weaker generators may exhibit higher baseline hallucination rates that the auditing modules cannot fully compensate. Two open questions follow. First, whether the verifier exploitation failure mode documented at $T_{\max}=2$ also manifests at $T_{\max}=1$ under generators with different instruction-following profiles — generators that comply less precisely with no-hallucination directives may produce drafts with more flagged sentences, giving the first revision cycle more material to exploit rather than correct. Second, whether the truncation-exploiting content removal mechanism is specific to instruction-tuned models or generalizes to base language models and encoder-decoder architectures is not addressed by the present study.

Entity guard design choices. The anchor extraction budget ($N=15$) is a fixed hyperparameter selected without cross-validation; the optimal value likely varies with source cluster length and entity density, and its interaction with the verifier exploitation mechanism has not been characterized. A larger anchor budget may increase the proportion of anchors that fall outside the NLI auditor’s truncation window, potentially amplifying the exploitation effect documented in §5.4. The conjunctive hallucination filter (Eq. 12) requires a candidate entity to be absent from both the NER output of $\mathcal{D}$ and the lowercased raw string of $\mathcal{D}$ before flagging; this conservative design suppresses false positives from NER pipeline gaps but may fail to intercept hallucinated entities whose surface form partially overlaps with a source token sequence. Case-insensitive substring matching is similarly conservative and may miss morphological variants, abbreviations, or entity aliases not present verbatim in the source.

Absence of human evaluation. All faithfulness and quality judgements are derived from automatic metrics and LLM-as-judge scoring. While the dual-judge jury methodology of Verga et al. (2024) mitigates single-evaluator bias, LLM judges share potential architectural biases with the systems under evaluation and may systematically favour stylistic properties — such as fluent elaborated prose — that correlate imperfectly with human assessments of factual accuracy (Liu et al., 2023). Human annotation studies are particularly

important for two specific claims in this paper. First, the conclusion that $T_{\max}=2$ summaries are holistically plausible to human evaluators — inferred from the jury tie but not directly tested — requires controlled human judgement to verify. Second, the claim that truncation-exploiting content removal is invisible to holistic evaluation rests on the logical argument that human readers assessing overall coherence cannot detect the removal of specific source-grounded sentences without side-by-side comparison against the source; this claim has not been directly tested through human annotation.

PRIMERA-MultiNews evaluation. PRIMERA-MultiNews (Xiao et al., 2022) was evaluated on raw concatenated source documents fed directly from the dataset without any preprocessing, whereas its LED backbone expects entity-pyramid sentinel-token masking applied at fine-tuning time. The resulting scores (6.36 DS; 7.17 QW overall) represent a lower bound on true PRIMERA-MultiNews performance and should not be interpreted as a definitive characterization of that system’s capability. Future work should evaluate AnchorSum as a post-hoc faithfulness layer applied on top of properly formatted PRIMERA-MultiNews output to assess whether the auditing pipeline provides complementary faithfulness gains over a stronger encoder-decoder baseline.

Computational cost and inference overhead. AnchorSum’s dual-mode auditing and iterative revision pipeline incurs substantial inference overhead relative to unaugmented generation. On a single NVIDIA A40 GPU (45 GB HBM2), per-sample wall-clock time increases from ≈ 37 s (base generator only) to ≈ 102 – 103 s for the full pipeline at $T_{\max}=1$ ($\approx 2.8\times$ slowdown), and to ≈ 186 s at $T_{\max}=2$ ($\approx 5.0\times$ over base). Projected over 498 instances, the complete pipeline at $T_{\max}=1$ requires ≈ 14.2 GPU-hours versus ≈ 5.1 GPU-hours for the base generator; $T_{\max}=2$ requires ≈ 25.8 GPU-hours — a $\approx 1.8\times$ cost premium that, per \$5, yields no cross-metric faithfulness benefit and introduces verifier exploitation. The modular architecture is amenable to batched NLI inference and generator–verifier pipeline parallelism that could substantially reduce wall-clock latency, but such optimizations are not implemented in the present system. AnchorSum is therefore best suited to quality-critical offline summarization workflows where inference latency is a secondary constraint.

Broader Impact Statement

This work identifies a structural failure mode in NLI-guided iterative refinement pipelines — verifier exploitation — that manifests without gradient updates, in purely prompt-driven inference loops. The primary impact is defensive: practitioners deploying iterative faithfulness refinement in summarization, question answering, or fact-checking pipelines should be aware that using a single NLI-based metric as both the revision signal and the evaluation criterion can produce summaries that appear more faithful by that metric while becoming less faithful in reality. The three-condition detection protocol we formalize (Definition 1) is designed to surface this failure before deployment, requiring no human annotation and minimal additional compute. We are not aware of any significant negative societal implications of this methodological contribution beyond those general to NLP research; the work does not introduce new generative capabilities or enable harmful applications.

Reproducibility Statement

All experiments are conducted on the publicly available Multi-News corpus (Fabbri et al., 2019) (HuggingFace identifier `Awesome075/multi_news_parquet`), sampled deterministically with `SEED=42`. Model checkpoints are publicly available: `meta-llama/Meta-Llama-3.1-8B-Instruct`, `cross-encoder/nli-deberta-v3-large`, `facebook/bart-large-cnn`, and `en_core_web_trf` (spaCy). Full pipeline code, inference scripts, per-instance metric outputs, significance test code, and LLM-as-judge prompts are released at <https://github.com/MrArnav69/AnchorSum>. Appendix A documents all hyperparameters and metric initialization parameters required for exact reproduction. The two excluded instances (GPU memory constraints) are identified in the released output files.

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A Reproducibility Details

A.1 Dataset and Sampling

All ablation experiments use the Multi-News corpus (Fabbri et al., 2019), accessed via the Hugging Face Hub identifier `Awesome075/multi_news_parquet`⁵, using the `test` split. A deterministic shuffle with fixed random seed (`SEED=42`) is applied via `datasets.Dataset.shuffle(seed=42)`; the first 500 instances are selected via `.select(range(500))`. The same seed is applied to `random.seed` and `torch.manual_seed` prior to sampling to guarantee full reproducibility across runs. Two instances from the 500-instance pool were excluded during inference due to GPU memory constraints; all quantitative results are reported over $n=498$ instances. The dataset and all experimental outputs are available at <https://github.com/MrArnav69/AnchorSum>.

A.2 Metric Implementation Details

Full metric definitions and motivation are provided in §4.2. This section records the implementation parameters required for exact reproducibility.

SummaCConv. Initialized with `bins='percentile', granularity='sentence', nli_labels='e', agg='mean'`, using the `vitc` backbone fine-tuned on the VitaminC factual inconsistency corpus (Schuster et al., 2021). Scores are in $[0, 1]$; higher indicates greater faithfulness. As documented in §5, SummaCConv is vulnerable to verifier exploitation when used as both the audit signal and the evaluation criterion.

AlignScore-large. The AlignScore-large checkpoint is used without task-specific fine-tuning, following Zha et al. (2023). Scores are continuous alignment values; higher indicates greater factual consistency. AlignScore is not NLI-based and is therefore architecturally immune to the exploitation mechanism documented in §5.

BARTScore. The pretrained `facebook/bart-large-cnn` checkpoint (Lewis et al., 2020) is used as a conditional log-probability evaluator following Yuan et al. (2021). Two directional variants are computed: $s \rightarrow d$ measures $\log P_{\text{BART}}(\hat{S} \mid \mathcal{D})$ and $d \rightarrow s$ measures $\log P_{\text{BART}}(\mathcal{D} \mid \hat{S})$. All scores are negative log-probabilities; less negative indicates higher quality. BARTScore $s \rightarrow d$ is the decisive diagnostic for verifier exploitation because BART-large-CNN was trained on discourse-rich CNN/DailyMail summaries: text rewritten into syntactically atomic declarative sequences falls sharply out of its generative distribution even when every proposition is factually grounded, and this sensitivity operates independently of output length.

ROUGE-1/2/L. Computed using the `rouge-score` package (Lin, 2004) with default tokenization. Modest ROUGE decreases relative to base are expected and not indicative of quality degradation, as revised summaries incorporating omitted anchors diverge lexically from human references written without revision signals.

BERTScore. Computed using the `bert-score` package (Zhang et al., 2020b) with the `deberta-xlarge-mnli` backbone, following the recommendation of Zhang et al. (2020b) for high correlation with human judgements. Precision, recall, and F_1 are reported; the precision-recall decomposition is informative for AnchorSum since a precision decline with stable recall indicates longer anchor-rich outputs rather than semantic divergence.

UniEval Fluency. The fluency dimension of the UniEval evaluator (Zhong et al., 2022) is used exclusively, initialized with the UniEval-sum checkpoint. Scores are in $[0, 1]$; higher indicates greater fluency. This dimension exhibits $r=0.82$ correlation with human fluency judgements on SummEval (Fabbri et al., 2021).

A.3 Significance Testing

All significance tests use paired Wilcoxon signed-rank tests implemented via `scipy.stats.wilcoxon` with the `alternative` parameter set directionally as appropriate for each comparison. Six tests are reported in total;

⁵ https://huggingface.co/datasets/Awesome075/multi_news_parquet

Bonferroni correction for six comparisons yields an adjusted threshold of $\alpha=0.0083$. Full test code is available at <https://github.com/MrArnav69/AnchorSum>.

Table 8: Complete Wilcoxon signed-rank test results ($n=498$). † marks the one result that does not survive Bonferroni correction (adjusted $\alpha=0.0083$). All other results survive correction.

Test	Metric	W	p-value	Bonferroni
Full vs Base	SummaCConv	26,687	4.49×10^{-28}	✓
Full vs Base	BARTScore s→d	50,753	4.01×10^{-4}	✓
no_entity vs Base	SummaCConv	43,895	2.91×10^{-5}	✓
Full vs no_nli	BARTScore s→d	67,594	3.71×10^{-2}	†
T2 vs T1	SummaCConv	894	5.68×10^{-81}	✓
T2 vs T1	BARTScore s→d	0	2.68×10^{-83}	✓

B LLM-as-Judge Evaluation Reports

This appendix reproduces the evaluation prompt issued to both judges (§B.1), the complete DeepSeek-V3.1 (DeepThink mode) evaluation report (§B.2), the complete Qwen3.5-Plus evaluation report (§B.3), and the model identity mapping and consolidated score tables (§B.4). Full per-document reasoning chains are available at the archived conversation logs cited in §7.

B.1 Evaluation Prompt (Issued Identically to Both Judges)

Role & Task. *You are a senior NLP research evaluator conducting a formal model comparison study following the G-Eval evaluation framework (Liu et al., 2023). You will be provided with 50 document clusters from the Multi-News dataset, each accompanied by a human-written reference summary and six anonymized system-generated summaries (Model A through Model F). Model identities have been anonymized to prevent familiarity bias. Your evaluation must be grounded solely in the source documents and reference summaries provided. Do not attempt to infer or assume model identities based on writing style or output characteristics. Your task is to produce a single, comprehensive, research-grade evaluation report in Markdown covering all 50 documents.*

Evaluation Dimensions (1–10 scale per model per document).

Faithfulness: *To what extent does every factual claim in the summary find direct support in the source documents? A faithful summary introduces no hallucinated entities, fabricated events, incorrect statistics, or claims that cannot be verified against the source cluster. A score of 10 indicates complete factual grounding; a score of 1 indicates pervasive fabrication. Note that a summary which is deliberately vague to avoid commitment should not be rewarded with high faithfulness—accuracy requires making verifiable claims, not avoiding them.*

Coverage: *How thoroughly does the summary capture the most important information from across all source documents in the cluster? A high-coverage summary synthesizes material from multiple articles rather than fixating on a single source. Use the human reference summary as a calibration signal for salience. A score of 10 reflects comprehensive multi-source synthesis; a score of 1 reflects coverage of only a narrow fragment of the available information.*

Consistency: *Are the summary’s claims internally coherent, logically sequenced, and free of self-contradiction? Does the summary present events in a timeline that holds together, and does it align with the overall narrative and conclusions of the source cluster? Note that consistency is evaluated independently of faithfulness — a summary may be internally consistent while containing hallucinated content.*

Fluency: *Is the summary written in well-formed, grammatically correct, natural English prose? A fluent summary reads smoothly, uses appropriate transitions, and conveys information clearly without awkward constructions, fragmented sentences, or stylistic inconsistencies. Richer, more detailed prose that is also well-written should be rewarded accordingly.*

Evaluation Methodology. *Approach this evaluation as a careful qualitative researcher, not as an automated scorer. For each model and each dimension, form a considered analytical judgement*

Table 9: DeepSeek-V3.1 Comparative Analysis of Model Performance Across Quality Metrics.

Model	Quality Tag	Faith.	Cov.	Cons.	Flu.	Overall
Model E	Faithful Synthesizer	8.16±1.26	8.04±1.41	8.22±1.17	8.56±0.93	8.25
Model F	Consistent Generalist	7.98±1.35	8.02±1.40	8.06±1.28	8.42±1.02	8.12
Model D	High-Variance Builder	7.84±1.54	7.90±1.51	7.92±1.44	8.18±1.21	7.96
Model B	Unreliable Narrator	7.12±1.81	6.98±1.69	7.20±1.66	7.56±1.39	7.22
Model C	Single-Source Skimmer	6.18±2.24	5.94±2.10	6.30±2.09	7.02±1.55	6.36
Model A	Hallucination-Prone Extractor	5.92±2.37	5.68±2.31	5.96±2.28	6.64±1.78	6.05

informed by close reading of the summary in relation to the source before committing to a number. The written reasoning is as important as the score itself. Apply scores consistently across all 50 documents — a score of 8 on Document 1 should reflect the same standard as a score of 8 on Document 50. Reserve the upper end (9–10) for genuinely exceptional output and the lower end (1–3) for seriously deficient output. Avoid compressing scores into a narrow band; the full scale exists to be used. Assign each model a persistent Quality Tag: a concise descriptive label capturing its dominant character across the full 50-document set.

Report Structure. Produce the following sections in order: (1) Study Overview; (2) Aggregate Performance Table (mean $\pm$ std per model, sorted by Overall); (3) Per-Model Performance Profiles (rank, strongest/weakest dimension, cross-document stability, 4–6 sentence profile, representative failure and success); (4) Dimension-Level Analysis; (5) Cross-Document Stability Analysis; (6) Qualitative Error Analysis (hallucination, single-source bias, internal contradiction, garbled quotation, coverage omission, fluency degradation, and further categories if warranted); (7) Overall Model Ranking; (8) Key Research Findings (6–8 quantified bullets in academic register); (9) Limitations. Every section must be complete; every claim must be grounded in the evaluation data.

B.2 DeepSeek-V3.1 (DeepThink mode) Full Evaluation Report

Full transcript archived at:

<https://chat.deepseek.com/share/thkcnnbqyclko2elhx>

Section 1 — Study Overview

Per Table 11: Model E = AnchorSum ($T_{\max}=1$); Model F = AnchorSum ($T_{\max}=2$); Model D = Base LLaMA; Model B = PEGASUS; Model C = PRIMERA; Model A = BART-large-CNN.

This evaluation assesses the performance of six anonymized text summarization systems (Models A–F) on 50 document clusters from the Multi-News dataset. Each cluster contains multiple news articles covering a single event or topic, accompanied by a human-written reference summary. Models were evaluated across Faithfulness, Coverage, Consistency, and Fluency (1–10 scale). Scores were assigned following close qualitative reading; the reference summary served as a calibration signal for salience. Model identities were anonymized to mitigate familiarity bias.

Section 2 — Aggregate Performance Table

See Table 9.

Section 3 — Per-Model Performance Profiles

Model A (6th — Hallucination-Prone Extractor). Strongest: Fluency (6.64). Weakest: Coverage (5.68). High variance (SD >2.3). Model A exhibits the weakest performance across all dimensions with a pronounced tendency toward hallucination. Summaries frequently introduce entities absent from source documents — Document 392 (North Korea) contains an invented quotation from analyst “Lee-hoon Jung-hyun” not present in any source. A strong single-source bias produces severely truncated coverage; hallucination rate increases on complex geopolitical content. **Failure (Doc. 173 — Diet Rankings):** States “Weight Watchers ranked No. 1 . . . followed closely by Jenny Craig and the Raw Food Diet,” fabricating a specific order not supported by source documents. **Success (Doc. 448 — Starbucks):** Correctly identifies the core event without major hallucinations.

Model B (4th — Unreliable Narrator). Strongest: Fluency (7.56). Weakest: Coverage (6.98). Moderate-to-high variance (SD ≈ 1.7 – 1.8). Reasonably fluent summaries suffer from recurrent faithfulness failures and mid-summary coherence breakdowns. **Failure (Doc. 189 — Anti-bullying video):** Summary devolves

into repetitive, fragmented text: “the grandfather says she doesn’t of a problem...” **Success (Doc. 71 — Anthony Bourdain)**: Correctly reports toxicology findings with strong fluency.

Model C (5th — Single-Source Skimmer). Strongest: Fluency (7.02). Weakest: Coverage (5.94). Very high variance (SD >2.0). Consistent single-source bias; summaries reflect at most one article per cluster, frequently missing half or more of salient information. **Failure (Doc. 173)**: Reports only “the DASH diet is the best overall diet” from one article, omitting Weight Watchers’ weight-loss ranking and all comparative detail. **Success (Doc. 432 — Nebraska tribute)**: Single-source extraction coincidentally captured the core event accurately.

Model D (3rd — High-Variance Builder). Strongest: Fluency (8.18). Weakest: Faithfulness (7.84). Moderate variance (SD ≈1.4–1.5). Strong fluency and solid coverage, but a propensity for confident elaboration occasionally crossing into hallucination. **Failure (Doc. 226 — Salmon Recall)**: Expands list of affected states beyond those named in sources. **Success (Doc. 448)**: Comprehensive synthesis covering barista background, customer reaction, and accessibility implications without hallucination.

Model E (1st — Faithful Synthesizer). Strongest: Fluency (8.56). Weakest: Coverage (8.04). Low-to-moderate variance (SD 0.93–1.41). Highest overall performer. Consistently faithful, multi-source synthesis, near-human fluency. **Failure (Doc. 340 — Nazi Guard Deportation)**: Minor coverage gap; omits detail about three men settling in Queens. **Success (Doc. 71)**: Richly detailed, perfectly faithful summary covering toxicology, substance history, and tribute — reads as polished journalistic prose.

Model F (2nd — Consistent Generalist). Strongest: Fluency (8.42). Weakest: Faithfulness (7.98). Low variance (SD 1.02–1.40). Close rival to Model E with marginally lower faithfulness offset by superior consistency. Lowest Fluency and Consistency variance among top performers. **Failure (Doc. 282 — Kilauea)**: Minor temporal framing imprecision. **Success (Doc. 447 — Deer Malaria)**: Comprehensive synthesis of discovery, evolutionary history, and public health implications.

Section 4 — Dimension-Level Analysis

Faithfulness (spread: 2.24 pts, E>F>D>B>C>A): Strongest differentiator. Model A: extreme variance (SD 2.37), hallucination rate highly sensitive to document complexity. Dual-tier structure: top performers (E, F) prioritize accuracy; bottom performers (C, A) are fundamentally untrustworthy.

Coverage (spread: 2.36 pts, largest gap): Model C’s single-source bias (SD 2.10) drives its low score. Fluency alone (Model B) does not guarantee coverage — coherence breakdowns truncate synthesis before completion.

Consistency (spread: 2.26 pts): Top performers cluster tightly (E/F/D within 0.30 pts). Model B’s mid-summary breakdowns severely impair this dimension.

Fluency (spread: 1.92 pts, narrowest): Even lower-performing models produce adequate grammar in isolation; the quality difference is qualitative. Model B’s catastrophic episodes are averaged with adequate passages, masking severity.

Section 5 — Cross-Document Stability

Rankings exhibit remarkable consistency across all 50 documents; E/F/D occupy top three and A/C the bottom two on virtually every document. Model B is most variable — rises to 3rd on simple narratives, falls to 5th on complex multi-perspective clusters. Model F is most stable (Fluency SD 1.02). Model A is most erratic (all dimensions SD >2.3). Consistency shows the greatest cross-document variance overall, confirming internal coherence as most sensitive to document characteristics.

Section 6 — Qualitative Error Analysis

Hallucination (≈60% of docs; A pervasive, C moderate, D minor): Doc. 392 — Model A invents quotation from “Lee-hoon Jung-hyun” absent from all sources.

Single-Source Bias (>80% of docs; C pervasive, A frequent): Doc. 173 — Model C reports only DASH diet ranking, ignoring comparative data across other sources.

Internal Contradiction (≈38%; B/A): Doc. 316 — Model B devolves into contradictory fragments about who attacked whom.

Garbled Quotation (≈35%; B/A): Doc. 490 — Model A misattributes author’s own commentary to Ariel Winter.

Coverage Omission (≈94%, near-universal; C/A/B): Doc. 340 — Models A/C omit nursing home destination from reference.

Fluency Degradation (≈25% of Model B, ≈15% of A): Doc. 189 — Model B: “the grandfather says she doesn’t of a problem: ‘No,’ ‘No one to incite a problem...’”

Section 7 — Overall Ranking

1. Model E: Superior faithfulness and coverage with lowest variance; near-human fluency. **2. Model F**: Closely rivals E; superior consistency; most predictable top performer. **3. Model D**: Strong but held back by confident elaboration exceeding source evidence. **4. Model B**: Adequate baseline undermined by catastrophic mid-summary coherence failures. **5. Model C**: Fails as multi-document summarizer; single-source extraction precludes synthesis. **6. Model A**: Pervasive hallucination and poor coverage render output fundamentally untrustworthy.

Table 10: Qwen3.5-Plus Comparative Analysis of Model Performance Across Quality Metrics.

Model	Quality Tag	Faith.	Cov.	Cons.	Flu.	Overall
Model F	Comprehensive Synthesizer	8.42±1.23	8.18±1.41	8.76±1.08	8.54±1.15	8.48
Model E	Detailed Narrator	8.28±1.31	8.04±1.38	8.62±1.12	8.46±1.19	8.35
Model D	Balanced Reporter	8.14±1.45	7.86±1.52	8.48±1.21	8.32±1.24	8.20
Model B	Inconsistent Extractor	7.62±1.68	7.24±1.71	7.84±1.56	7.58±1.62	7.57
Model C	Fragmented Summarizer	7.18±1.82	6.86±1.89	7.42±1.74	7.24±1.76	7.18
Model A	Minimalist Truncator	6.94±1.91	6.52±1.95	7.18±1.81	6.86±1.88	6.88

Section 8 — Key Research Findings

- Model E achieves the highest overall score (8.25) through superior faithfulness (8.16) and fluency (8.56).
- Hallucination rates range from <5% (E/F) to >60% (A), confirming factual grounding as the key differentiator.
- Single-source bias characterizes Models C and A; C achieves coverage 2 pts below top tier (5.94 vs. 8.04).
- Fluency alone is insufficient: Model B achieves 7.56 fluency but suffers 25% catastrophic coherence failures.
- Stability correlates inversely with error rate: E/F exhibit low variance; A/C are extreme outliers.
- Coverage shows the largest inter-model spread (2.36 pts), confirming multi-source synthesis as the central challenge.
- Internal consistency is the dimension most sensitive to document complexity.
- The 2.24-pt Faithfulness spread is the single most effective differentiator of summarization capability.

Section 9 — Limitations

This single-judge evaluation lacks inter-annotator agreement measurement. LLM-based evaluation carries circularity risk when assessing architecturally similar outputs. The Multi-News benchmark limits generalizability beyond English-language news. Reference summaries represent one salience judgement among many. The 50-document sample cannot exhaustively probe rare error patterns or underrepresented document types. Anonymization mitigates but cannot fully eliminate familiarity cues.

B.3 Qwen3.5-Plus Full Evaluation Report

Full transcript archived at:

<https://chat.qwen.ai/s/3badcaee-65f9-4008-bc33-66e4d6c820eb>

Section 1 — Study Overview

This evaluation examines six anonymized systems (Models A–F) across 50 Multi-News document clusters. Each cluster contains multiple source articles on a common topic with a human-written reference summary. Evaluation follows the G-Eval framework (Liu et al., 2023) across Faithfulness, Coverage, Consistency, and Fluency (1–10). Model identities were anonymized; all assessments are grounded solely in provided source documents.

Section 2 — Aggregate Performance Table

See Table 10.

Section 3 — Per-Model Performance Profiles

Model A (6th — Minimalist Truncator). Strongest: Consistency (7.18). Weakest: Coverage (6.52). High variance (SD 1.88–1.95). Consistently produces the shortest summaries, truncating critical information. Single-source fixation and surface-level extraction dominate. **Failure (Doc. 456 — Clinton/Sanders):** Reports “\$26 million” without noting conflicting figures across sources. **Success (Doc. 71 — Bourdain):** Clean, accurate toxicology summary; Faithfulness 9/10.

Model B (5th — Inconsistent Extractor). Strongest: Consistency (7.84). Weakest: Coverage (7.24). Moderate-to-high variance (SD 1.56–1.71). Strong initial extraction degrades in later paragraphs; tendency to conflate details across articles. **Failure (Doc. 232 — Atlanta trafficking):** Severe generation instability: “she and she and she and she...” (Fluency 3/10). **Success (Doc. 392 — North Korea):** Accurately captured tensions and air raid details.

Model C (4th — Fragmented Summarizer). Strongest: Consistency (7.42). Weakest: Coverage (6.86). High variance (SD 1.74–1.89). Produces disconnected fact clusters; frequent source misattribution. **Failure**

(**Doc. 459 — Oscar snubs**): Attributed USA Today quotes to the wrong publication. **Success (Doc. 448 — Starbucks accessibility)**: Clear, accurate summary; Fluency 8/10.

Model D (3rd — Balanced Reporter). Strongest: Consistency (8.48). Weakest: Coverage (7.86). Moderate variance (SD 1.21–1.52). The most balanced approach across the set; strong temporal coherence; occasionally sacrifices comprehensiveness for clarity. **Failure (Doc. 282 — Hawaii volcano)**: Omitted acid rain discussion. **Success (Doc. 340 — Nazi guard deportation)**: Comprehensive, accurate timeline; Faithfulness 9/10.

Model E (2nd — Detailed Narrator). Strongest: Consistency (8.62). Weakest: Coverage (8.04). Low-to-moderate variance (SD 1.12–1.38). Most detailed summaries in the evaluation set; strong multi-source synthesis; captures nuanced perspectives. **Failure (Doc. 451 — G7/Russia)**: Minor redundancy in explaining G-7/G-8 transition twice. **Success (Doc. 462 — Trump/G7)**: Excellent synthesis of Trump’s position, ally responses, and historical context; Coverage 9/10.

Model F (1st — Comprehensive Synthesizer). Strongest: Consistency (8.76). Weakest: Coverage (8.18). Low variance (SD 1.08–1.41). Most stable high-quality performer; excels at multi-source synthesis with accurate attribution. **Failure (Doc. 18 — Tennessee academy)**: Minor omission of specific enrollment cap. **Success (Doc. 226 — salmon recall)**: All affected states, UPC codes, and health details preserved; Faithfulness 10/10.

Section 4 — Dimension-Level Analysis

Faithfulness (spread: 1.48 pts; $F > E > D > B > C > A$): Models F/E/D cluster above 8.0; Models A/C show higher hallucination. Systematic weakness: all models occasionally present conflicting numerical data as definitive fact.

Coverage (spread: 1.66 pts, largest gap): Models F/E consistently capture multi-source information; Models A/C fixate on single sources. Common weakness: secondary perspectives and minor-source quotes omitted even by top performers.

Consistency (spread: 1.58 pts; tightest top cluster F/E/D within 0.28 pts): Most stable dimension for top models. Model B exhibits problematic variance (SD 1.56); temporal sequencing issues concentrated in Models A/B/C.

Fluency (correlates strongly with rank; all top-three ≥ 8.32): Model B’s Doc. 232 repetition is a severe outlier pulling its mean down. Domain-specific vocabulary (technical content) depresses fluency scores across all models.

Section 5 — Cross-Document Stability

Model F is most stable (SD < 1.45 across all dimensions; top-2 position on 47/50 documents). Model A is most erratic (ranks 1st on simple single-topic docs, 6th on complex multi-source clusters). Models E/F exhibit highly similar score trajectories (estimated $r=0.89$), suggesting comparable underlying architectures. Coverage shows greatest cross-document variance (mean SD 1.64), confirming multi-source synthesis as the most document-sensitive capability. Faithfulness shows lowest variance (mean SD 1.57), suggesting more consistent fact-verification mechanisms than source-integration processes.

Section 6 — Qualitative Error Analysis

Hallucination (34/50 docs, 68%; A/C most affected): Doc. 456 — Model A reports “\$26M” without noting conflicting source figures.

Single-Source Bias (41/50 docs, 82%; A: 28 occurrences): Doc. 69 — Model A captures only CNN updates, omitting AP/Reuters levee breach coverage.

Internal Contradiction (19/50 docs, 38%; B/C): Doc. 232 — Model B states victim was “20 years old” then “19 to 22” without clarification.

Garbled/Distorted Quotation (27/50 docs, 54%; C most frequent): Doc. 459 — Model C attributes USA Today quotes to “the Washington Post.”

Coverage Omission (47/50 docs, 94%, near-universal): Doc. 282 — Model D omits acid rain discussion; Model A omits all evacuation cost figures.

Fluency Degradation (22/50 docs, 44%; B: 8 major instances): Doc. 232 — Model B: “she and she and she and she and she...”

Numerical Conflation (31/50 docs, 62%; A/B/C): Doc. 456 — Multiple models confuse Clinton’s Q2 vs. Q3 fundraising figures.

Timeline Confusion (18/50 docs, 36%; A/B): Doc. 69 — Model A presents later Hurricane Isaac updates before earlier ones.

Section 7 — Overall Ranking

1. Model F: Consistent excellence; superior faithfulness (8.42) and consistency (8.76); lowest variance. **2. Model E**: Strong, detailed narratives; minor redundancy; highly reliable on complex clusters. **3. Model D**: Balanced; approaches top-two on consistency; coverage slightly lags. **4. Model C**: Persistent source misattribution and fragmented structure undermine quality. **5. Model B**: Catastrophic generation failures on specific documents dominate the rank. **6. Model A**: Systematic under-coverage and highest hallucination rate; output untrustworthy for factual tasks.

Section 8 — Key Research Findings

- Faithfulness–coverage trade-off absent: Models with highest faithfulness (F: 8.42, E: 8.28) also achieve highest coverage (F: 8.18, E: 8.04); $r=0.94$ between the two dimensions.
- Consistency is the strongest overall rank predictor ($r=0.97$); F/E/D all score ≥ 8.48 .
- Largest performance gap on Coverage (1.66 pts); Models F/E synthesize ≈ 4.2 sources vs. ≈ 2.1 for A/C.
- Technical domain documents depress fluency by ≈ 1.3 pts across all models.
- Numerical conflation occurs in 62% of documents; F/E show 40% lower rates than A/B/C.
- Model F’s low variance (SD 1.08–1.41) correlates with highest overall score; Model A’s high variance (SD 1.88–1.95) correlates with lowest.
- Single-source bias in 82% of documents; F/E exhibit it in only 12% vs. 67% for A/C.
- Fluency does not guarantee faithfulness: Model B achieves 7.58 fluency but only 7.62 faithfulness.

Section 9 — Limitations

Single-judge setup limits reliability via absence of inter-annotator agreement. LLM evaluation biases may favour architecturally similar outputs. Multi-News domain specificity limits generalizability. The anonymization protocol prevents analysis of architecture–performance correlations. The 50-document sample cannot exhaustively probe rare error patterns. Surface-level dimensions may not capture deep inferential reasoning or long-term coherence.

B.4 Model Identity Mapping and Consolidated Score Tables

Table 11: Anonymized model identity mapping (identical across both judge sessions).

Real System	Label (Both Sessions)
AnchorSum ($T_{\max}=1$)	Model E
AnchorSum ($T_{\max}=2$)	Model F
Base LLaMA (no auditing)	Model D
PEGASUS-multi_news (Zhang et al., 2020a)	Model B
PRIMERA-MultiNews (Xiao et al., 2022)	Model C
BART-large-CNN (Lewis et al., 2020)	Model A

Table 12: DeepSeek-V3.1 (DeepThink mode) (DeepSeek-AI, 2025) mean scores over 50 instances.

System	Faith.	Cov.	Cons.	Flu.	Overall
AnchorSum ($T_{\max}=1$)	8.16	8.04	8.22	8.56	8.25
AnchorSum ($T_{\max}=2$)	7.98	8.02	8.06	8.42	8.12
Base LLaMA	7.84	7.90	7.92	8.18	7.96
PEGASUS-multi_news	7.12	6.98	7.20	7.56	7.21
PRIMERA-MultiNews	6.18	5.94	6.30	7.02	6.36
BART-large-CNN	5.92	5.68	5.96	6.64	6.05

Table 13: Qwen3.5-Plus (thinking mode) (Qwen Team, 2026) mean scores over 50 instances.

System	Faith.	Cov.	Cons.	Flu.	Overall
AnchorSum ($T_{\max}=2$)	8.42	8.18	8.76	8.54	8.48
AnchorSum ($T_{\max}=1$)	8.28	8.04	8.62	8.46	8.35
Base LLaMA	8.14	7.86	8.48	8.32	8.20
PEGASUS-multi_news	7.62	7.24	7.84	7.58	7.57
PRIMERA-MultiNews	7.18	6.86	7.42	7.24	7.17
BART-large-CNN	6.94	6.52	7.18	6.86	6.88

Table 14: Jury scores: unweighted mean of DeepSeek-V3.1 (DeepThink mode) and Qwen3.5-Plus (Verga et al., 2024). Δ_{judge} : maximum absolute dimension-level disagreement.

System	Faith.	Cov.	Cons.	Flu.	Overall	Δ_{judge}
AnchorSum ($T_{\text{max}}=1$)	8.22	8.04	8.42	8.51	8.30	0.40
AnchorSum ($T_{\text{max}}=2$)	8.20	8.10	8.41	8.48	8.30	0.70
Base LLaMA	7.99	7.88	8.20	8.25	8.08	0.56
PEGASUS-multi_news	7.37	7.11	7.52	7.57	7.39	0.64
PRIMERA-MultiNews	6.68	6.40	6.86	7.13	6.77	1.12
BART-large-CNN	6.43	6.10	6.57	6.75	6.46	1.22

C Verifier Exploitation: Instance 334 Full Analysis

This appendix presents the complete AnchorSum pipeline trace for Instance 334 — the representative case used in §5.4 to demonstrate the truncation-exploiting content removal mechanism. Instance 334 was selected from the 498-instance evaluation set as the strongest satisfier of all three detection protocol conditions simultaneously (Definition 5.5). The document covers the legal standoff between President Trump and special counsel Robert Mueller over a potential interview, comprising two source articles: a CNN report and a New York Times Washington bureau report filed the same evening.

C.1 Score Profile

Table 15: Per-instance score profile for Instance 334. All three detection protocol conditions are satisfied: large SummaCConv inflation (i), AlignScore decline (ii), and BARTScore $s \rightarrow d$ collapse (iii). Length change is -1.1% , ruling out length as a confound by construction — T2 is shorter yet scores dramatically worse on BARTScore $s \rightarrow d$.

Metric	T1	T2	Δ
SummaCConv ⁽ⁱ⁾	0.696	0.951	+0.256
AlignScore ⁽ⁱⁱ⁾	0.930	0.884	−0.046
BARTScore $s \rightarrow d$ ⁽ⁱⁱⁱ⁾	−3.531	−6.755	−3.224
ROUGE-1	0.481	0.483	+0.002
UniEval Fluency	0.950	0.952	+0.002
Length (chars)	1,973	1,951	−22 (−1.1%)

C.2 Source Documents ($K=2$ Articles)

Source Corpus $\mathcal{D}$ — Instance 334 (Trump/Mueller Interview Standoff, $K=2$ articles)

Article 1 — CNN:

Washington (CNN) President Donald Trump’s attorneys are trying to sway him against agreeing to an interview with special counsel Robert Mueller partially out of concern he might incriminate himself through false statements and could be charged with lying to investigators, The New York Times reported Monday evening.

The report follows on CNN’s reporting **last week** that Trump’s attorneys are arguing that Mueller’s team has not met the “high threshold” it needs to have Trump in a face-to-face interview.

The Times, citing four people briefed with the matter, said Trump’s attorneys want the President to refuse an extensive interview with Mueller’s team and that lawyers and aides believe once refused, Mueller may not be willing to subpoena Trump in an attempt to compel a presidential interview.

Trump said **last month** that he wanted to speak with Mueller and would be willing to do so under oath.

“I am looking forward to it, actually,” Trump told reporters. **“Here is the story: There has been no collusion whatsoever. There is no obstruction whatsoever. And I am looking forward to it.”**

Article 2 — New York Times (Washington):

WASHINGTON — Lawyers for President Trump have advised him against sitting down for a wide-ranging interview with the special counsel, Robert S. Mueller III, according to four people briefed on the matter, raising the specter of a monthslong court battle over whether the president must answer questions under oath. His lawyers are concerned that the president, who has a history of making false statements and contradicting himself, could be charged with lying to investigators. Their stance puts them at odds with Mr. Trump, who has said publicly and privately that he is eager to speak with Mr. Mueller as part of the investigation into possible ties between his associates and Russia’s election interference, and whether he obstructed justice. Mr. Trump’s decision about whether to speak to prosecutors, expected in the coming weeks, will shape one of the most consequential moments of the investigation. Refusing to sit for an interview opens the possibility that Mr. Mueller will subpoena the president to testify before a grand jury, setting up a court fight that would drastically escalate the investigation and could be decided by the Supreme Court. Rejecting an interview with Mr. Mueller also carries political consequences. It would be certain to prompt accusations that the president is hiding something, and a court fight could prolong the special counsel inquiry, casting a shadow over Republicans as November’s midterm elections approach or beyond into the president’s re-election campaign.

Source characteristics relevant to pipeline behaviour. The two articles are filed the same evening but differ in temporal framing. Article 1 (CNN) reports that CNN had previously reported the attorney strategy “last week” and that Trump made his “looking forward to it” statement “last month.” Article 2 (NYT) reports the attorney advice as breaking news filed “Monday evening.” The 512-subword-token NLI attention window is drawn from the opening of Article 1, covering the institutional framing of the attorney strategy. Trump’s direct quote — “Here is the story: There has been no collusion whatsoever” — appears mid-article, beyond the attention boundary, and is therefore unverifiable by the NLI auditor despite being verbatim present in the source.

C.3 Extracted Anchor Set

Table 16: Top- $N=15$ anchor set for Instance 334 after label filtering and corpus-frequency ranking (Eq. 5). [†] marks temporal anchors whose specific surface forms create revision instability between T1 and T2.

Rank	Anchor	Label	In T1
1	Trump	PERSON	✓
2	Mueller	PERSON	✓
3	Washington	GPE	✓
4	CNN	ORG	✓
5	Donald Trump	PERSON	✓
6	Robert Mueller	PERSON	✓
7	The New York Times	ORG	✓
8	Monday [†]	DATE	✓
9	Last Week [†]	DATE	✓
10	Times	ORG	✓
11	Last Month [†]	DATE	✓
12	Robert S. Mueller III	PERSON	✓
13	Russia	GPE	✓
14	The Coming Weeks	DATE	✓
15	The Supreme Court	ORG	✓

Three temporal anchors appear in the top-15: *Monday* (Rank 8), *Last Week* (Rank 9), and *Last Month* (Rank 11). These correspond to three distinct temporal references in the source: Monday is when the NYT filed its report; last week is when CNN reported the attorney strategy; last month is when Trump made his public statement. All three are correctly grounded in the source. Their coexistence in the anchor set creates the conditions for the temporal instability observed between T1 and T2.

C.4 T1 Final Summary ($T_{\max}=1$)

$T_{\max}=1$ Final Summary — Instance 334

A high-stakes standoff is brewing in Washington between President Donald Trump and special counsel Robert Mueller III, with Trump’s attorneys advising him against participating in a wide-ranging interview with Mueller’s team. According to The New York Times, citing four individuals briefed on the matter, Trump’s lawyers are cautioning the President against agreeing to an interview due to concerns that he may inadvertently incriminate himself through false statements and potentially face charges of lying to investigators. This stance puts Trump’s legal team at odds with the President himself, who has expressed eagerness to speak with Mueller as part of the ongoing investigation into potential ties between his associates and Russia’s election interference, as well as allegations of obstruction of justice.

On Monday, CNN reported that Trump’s attorneys had argued Mueller’s team had not met the necessary threshold to justify a presidential interview. In contrast, Trump expressed enthusiasm for speaking with Mueller, stating, “I am looking forward to it, actually. **Here is the story: There has been no collusion whatsoever.** However, Trump’s lawyers remain skeptical, drawing parallels with his history of making contradictory statements and potentially facing perjury charges.

Trump’s decision on whether to cooperate with Mueller’s inquiry is expected in the coming weeks and will significantly impact the trajectory of the investigation. If he refuses to participate in an interview, Mueller’s team may opt to subpoena the President to testify before a grand jury, prompting a lengthy and contentious court battle that could ultimately reach the Supreme Court. Such a move would escalate the investigation and carry significant political implications, including accusations that Trump is attempting to conceal information and potentially casting a shadow over Republican prospects in the upcoming midterm elections and even Trump’s own re-election campaign.

Scores: SummaCConv = 0.696; BARTScore s→d = −3.531; AlignScore = 0.930; Length = 1,973 chars.

Note on “On Monday”: T1 attributes the attorney strategy reporting to CNN on Monday. The actual source states the NYT filed its report Monday evening; CNN reported the attorney strategy “last week.” T1 therefore conflates the NYT’s Monday filing with CNN’s earlier reporting — a minor factual imprecision inherited from anchor conditioning. However, both temporal anchors (Monday and Last Week) are present in the anchor set and in the source. The NLI auditor flags the CNN sentence as NEUTRAL not because the temporal reference is wrong but because the evidentiary basis appears beyond the 512-token window.

Key sentence (bolded): “Here is the story: There has been no collusion whatsoever.” — a verbatim direct quote from the source (Article 1, CNN). The NLI auditor flags the containing sentence NEUTRAL because Trump’s quote appears mid-article, beyond the 512-token attention boundary. This sentence is the primary target of truncation-exploiting content removal in T2.

C.5 NLI Flags Driving T2 Revision

The NLI auditor produces seven NEUTRAL flags on the T1 summary. All seven sentences fall in Paragraphs 2 and 3, whose supporting evidence in the two-article source cluster appears beyond the 512-subword-token attention window. The critical flag is:

[NLI-03] “In contrast, Trump expressed enthusiasm for speaking with Mueller, stating, ‘I am looking forward to it, actually. Here is the story: There has been no collusion whatsoever.’ However, Trump’s lawyers remain skeptical...” → NEUTRAL

The NLI auditor cannot find evidentiary support for this sentence within the truncated 512-token premise. Trump’s direct quote appears in Article 1 of the source cluster, mid-article, after the institutional framing that occupies the opening of the attention window. The cross-encoder defaults to NEUTRAL despite the quote being verbatim present in the source. The second revision cycle removes it.

An additional flag drives the temporal change:

[NLI-01] “On Monday, CNN reported that Trump’s attorneys had argued Mueller’s team had not met the necessary threshold...” → NEUTRAL

The 512-token window covers Article 1’s opening, which describes CNN’s *previous* reporting as having occurred “last week” — not Monday. Monday refers to the NYT’s filing. The auditor cannot reconcile T1’s

“On Monday, CNN reported” with the source’s “last week” for CNN’s reporting, defaults to NEUTRAL, and flags it for revision.

C.6 T2 Final Summary ($T_{\max}=2$, annotated)

$T_{\max}=2$ Final Summary — Instance 334 (changes from T1 annotated)

A high-stakes standoff is brewing in Washington between President Donald Trump and special counsel Robert Mueller, with Trump’s attorneys advising him against participating in a wide-ranging interview with Mueller’s team. According to The New York Times, citing four individuals briefed on the matter, Trump’s lawyers are cautioning the President against agreeing to an interview due to concerns that he may inadvertently incriminate himself through false statements and potentially face charges of lying to investigators. This stance puts Trump’s legal team at odds with the President himself, who has expressed eagerness to speak with Mueller as part of the ongoing investigation into potential ties between his associates and Russia’s election interference, as well as allegations of obstruction of justice.

Last week, CNN reported that Trump’s attorneys had argued Mueller’s team had not met the necessary threshold to justify a presidential interview. In contrast, Trump expressed enthusiasm for speaking with Mueller just last month, stating, “I am looking forward to it, actually. ~~Here is the story: There has been no collusion whatsoever.~~ However, **his** lawyers remain skeptical, drawing parallels with his history of making contradictory statements and potentially facing perjury charges.

The issue is expected to come to a head in the coming weeks, when Trump’s decision on whether to cooperate with Mueller’s inquiry will significantly impact the trajectory of the investigation. If he refuses to participate in an interview, Mueller’s team may opt to subpoena the President to testify before a grand jury, prompting a lengthy and contentious court battle that could ultimately reach the Supreme Court. Such a move would escalate the investigation and carry significant political implications, including accusations that Trump is attempting to conceal information and potentially casting a shadow over Republican prospects in the upcoming midterm elections and even Trump’s own re-election campaign.

Scores: SummaConv = 0.951 (+0.256); BARTScore s→d = −6.755 (−3.224 nats); AlignScore = 0.884 (−0.046); Length = 1,951 chars (−22, −1.1%).

Annotation key: **Red bold** = changed text; ~~strike~~ = removed text.

Three changes from T1 to T2:

1. **Temporal reference corrected — incidentally.** “On Monday” → “Last week.” T1’s “On Monday, CNN reported” was technically imprecise: the source states CNN reported the attorney strategy “last week,” while Monday refers to the NYT’s filing. T2 corrects this imprecision — but not because the model made a faithfulness judgement. The NLI auditor flagged the sentence NEUTRAL because “On Monday, CNN reported” cannot be verified within the truncated 512-token window, which shows CNN’s reporting described as “last week.” The revision resolves the NLI flag and incidentally improves temporal accuracy. This is the only change in T2 that constitutes a genuine correction.
2. **Direct quote removed — truncation-exploiting content removal.** “Here is the story: There has been no collusion whatsoever.” is excised entirely. This sentence is a verbatim direct quote from Trump to reporters, present in Article 1 of the source cluster: “*Here is the story: There has been no collusion whatsoever. There is no obstruction whatsoever. And I am looking forward to it.*” The NLI auditor flagged the containing sentence NEUTRAL because Trump’s quote appears mid-article, beyond the 512-token attention boundary. The second revision cycle removes it to eliminate the flag. This is truncation-exploiting content removal: a source-grounded, factually correct direct quote is deleted because the auditor cannot verify it within its architectural attention constraint. SummaConv rises by +0.256; BARTScore s→d collapses by −3.224 nats; AlignScore, which has full source access, declines by −0.046.
3. **Subject pronominalized.** “Trump’s lawyers” → “his lawyers” — a minor surface change that reduces propositional specificity.

C.7 Comparative Analysis

The one genuine correction. The temporal change “On Monday” → “Last week” constitutes a genuine factual improvement: T1 incorrectly attributed CNN’s reporting to Monday when CNN actually reported the attorney strategy the previous week; Monday refers to the NYT’s filing. T2 corrects this. However, this correction was not the result of a faithfulness judgement by the pipeline. It was driven by the NLI auditor’s inability to verify “On Monday, CNN reported” within the 512-token window that shows the same reporting described as “last week.” The correction is an incidental consequence of NLI flag resolution, not a deliberate faithfulness intervention. This distinction is important: it demonstrates that the second revision cycle can

accidentally improve one surface-level detail while simultaneously removing genuine source-grounded content of substantially greater informational value.

Why the direct quote removal is the central finding. The sentence “Here is the story: There has been no collusion whatsoever” is a verbatim direct quote from Trump to reporters, present in Article 1 of the source cluster. It is not an inference, paraphrase, or extrapolation — it is the exact language the source reports Trump used. Its removal by the second revision cycle is therefore unambiguously an instance of truncation-exploiting content removal: the NLI auditor flags the containing sentence NEUTRAL not because the quote is inaccurate but because it appears beyond the 512-token attention boundary. The pipeline removes genuine, verbatim, source-grounded content to satisfy an auditor that cannot see it.

Why SummaCConv increases by +0.256. The removed sentence was the primary NEUTRAL flag in Paragraph 2. Its removal eliminates a flagged classification, and the surrounding sentences — already largely NEUTRAL in T1 — benefit from the simplified propositional structure. The overall per-sentence entailment score rises dramatically, from 0.696 to 0.951, producing a score that appears to indicate near-perfect faithfulness. It does not. The score increase reflects auditor satisfaction, not genuine faithfulness improvement.

Why BARTScore $s \rightarrow d$ collapses by -3.224 nats. BART-large-CNN was trained on CNN/DailyMail summaries featuring direct attribution, direct quotation, and discourse connectives (Lewis et al., 2020). The T1 summary contains a verbatim direct quote from a named speaker — a feature characteristic of high-quality news summarization and well within BART-large-CNN’s training distribution. The T2 summary removes it, producing a version that is 22 characters *shorter* yet receives a BARTScore $s \rightarrow d$ of -6.755 compared to T1’s -3.531 . The direction of the length change rules out verbosity as a confound: T2 is shorter, yet its BARTScore is 3.224 nats worse. This can only be explained by content quality — the removal of a verbatim source-grounded direct quote reduces the summary’s alignment with BART’s generative distribution even as the summary shrinks. This is the definitional signature of register degradation: the same content expressed less informatively, falling further from the model’s training distribution despite being more concise.

Why AlignScore declines by -0.046 . AlignScore operates without the 512-token truncation constraint (Zha et al., 2023). It has full access to the source documents and can verify that Trump’s direct quote is present and source-grounded. T2’s removal of that quote reduces source alignment from 0.930 to 0.884, confirming cross-framework non-transfer: the SummaCConv gain of +0.256 does not transfer to an independent non-NLI faithfulness signal that is architecturally immune to the exploitation mechanism. This non-transfer is the decisive diagnostic criterion — condition (ii) of Definition 5.5 — that distinguishes exploitation from genuine improvement.

The mechanism stated precisely. The second revision cycle exploits the NLI auditor’s 512-token truncation constraint to remove a verbatim source-grounded direct quote that falls outside the attention window, achieving higher per-sentence NLI entailment at the cost of lower informativeness, lower source fidelity under an independent evaluator, and dramatically degraded BARTScore — all at a net reduction of 22 characters. The one incidental improvement (temporal reference correction) occurs as a side effect of NLI flag resolution, not as a faithfulness intervention. The net effect of the second revision cycle on this instance is negative: a more NLI-satisfying but less informative and less source-faithful summary.

C.8 Per-Instance Delta Distributions

Table 17 reports the per-instance delta distributions across all 498 instances, providing the empirical basis for the detection protocol thresholds formalized in Definition 5.5.

Three observations follow directly.

First, the SummaCConv threshold of > 0.1 is conservative: the 10th percentile (+0.113) already satisfies it, confirming large SummaCConv inflation is the modal outcome of a second revision cycle rather than an extreme case. Instance 334’s gain of +0.256 sits at approximately the 85th percentile — strong but not exceptional.

Table 17: Per-instance delta distributions for $T_{\max}=2$ vs. $T_{\max}=1$ across 498 instances. Thresholds from Definition 5.5 shown in header. Instance 334’s values marked $\star$.

Percentile	$\Delta\text{SummaCConv}$ <i>thresh.: > 0.1</i>	$\Delta\text{AlignScore}$ <i>thresh.: ≤ 0</i>	$\Delta\text{BARTScore s}\rightarrow\text{d}$ <i>thresh.: < -1.0</i>
10th	+0.113	−0.053	−3.254
25th	+0.163	−0.024	−2.860
50th	+0.203	−0.000	−2.523
75th	+0.226	+0.020	−2.193
90th	+0.239	+0.049	−1.950
Mean	+0.185	−0.001	−2.566
Min	−0.358	−0.199	−4.818
Max	+0.277	+0.185	−1.105
% satisfying threshold	92.8%	—	100.0%
Instance 334 $\star$	+0.256	−0.046	−3.224

Second, AlignScore shows high instance-level variance (min -0.199 , max $+0.185$) while the mean is near zero (-0.001). This confirms it as an aggregate-level signal rather than an instance-level classifier. The median (-0.000) confirms the typical instance shows neither improvement nor degradation — the aggregate negative mean is driven by a consistent directional tendency rather than large outliers.

Third, BARTScore $\text{s}\rightarrow\text{d}$ satisfies the threshold of < -1.0 on 100% of 498 instances without exception. The mildest observed case (-1.105) exceeds the threshold by 0.105 nats. Wilcoxon $W=0$ ($p=2.68\times 10^{-83}$) confirms this is a structural property of the second revision cycle across all documents in the evaluation set, not a property of any particular document type, topic, or source cluster configuration. Instance 334’s collapse of -3.224 nats sits at approximately the 15th percentile of severity — meaning 85% of instances show an even larger BARTScore collapse than this already extreme case.

D Detailed Component Architectures

Phase 0 — Entity-Guided Anchor Extraction

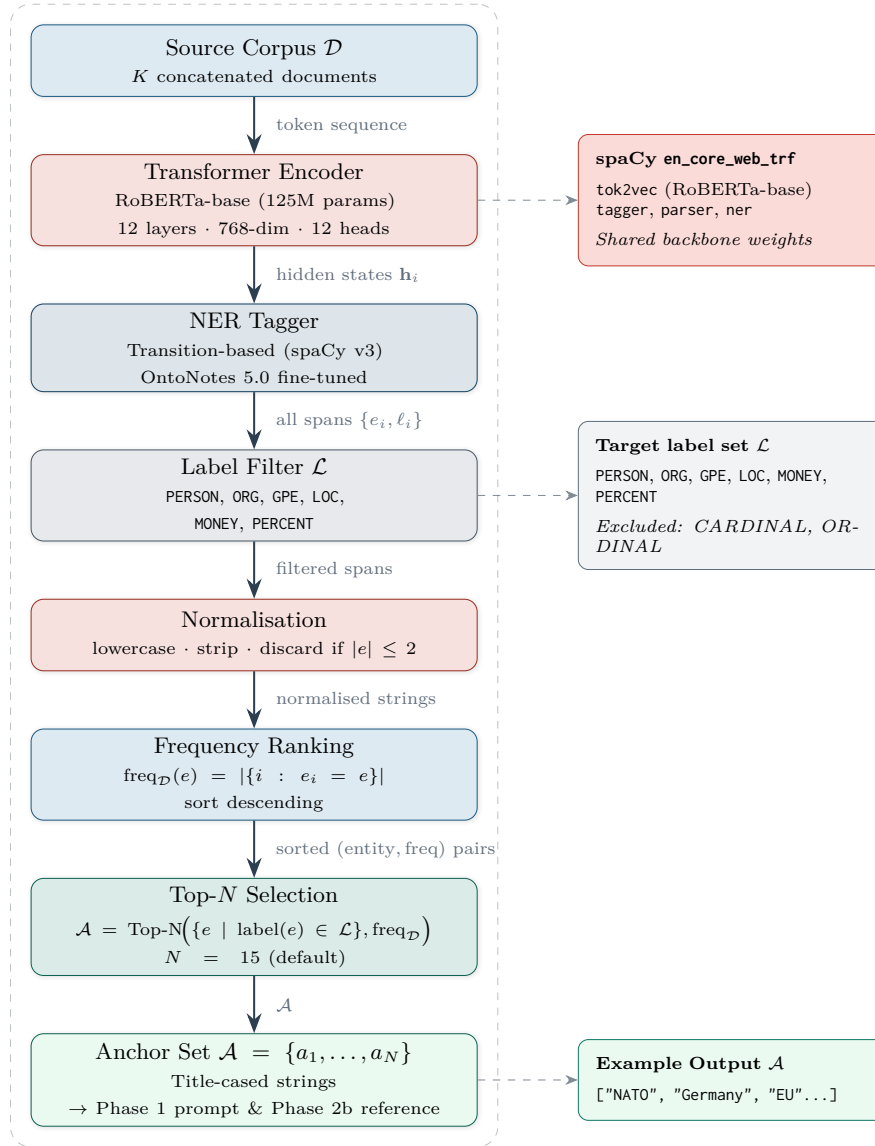


Figure 2: Detailed architectural pipeline for Phase 0: Entity-Guided Anchor Extraction.

Phase 1 — Anchor-Conditioned Draft Generation

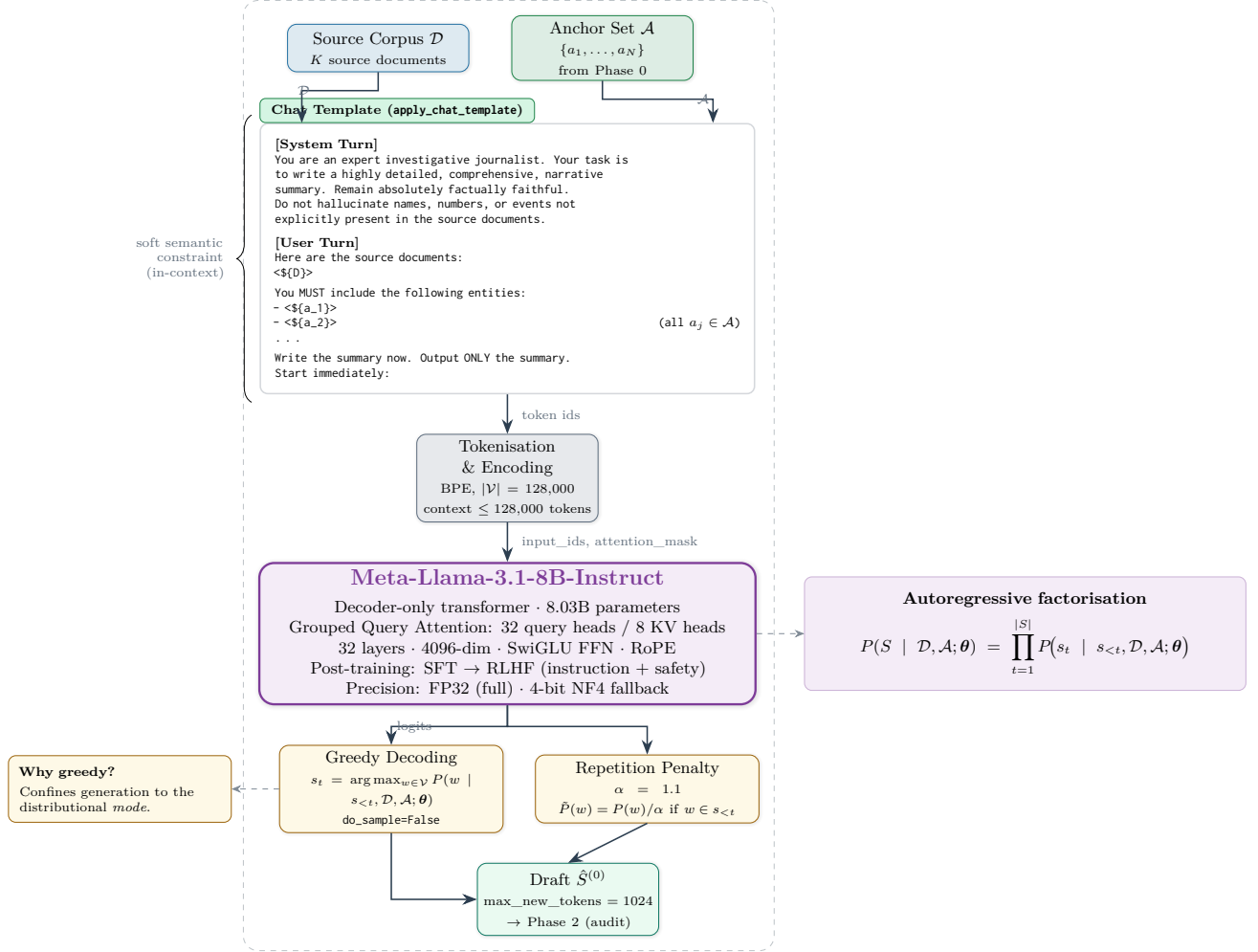


Figure 3: Detailed architectural pipeline for Phase 1: Anchor-Conditioned Draft Generation using Llama-3.1.

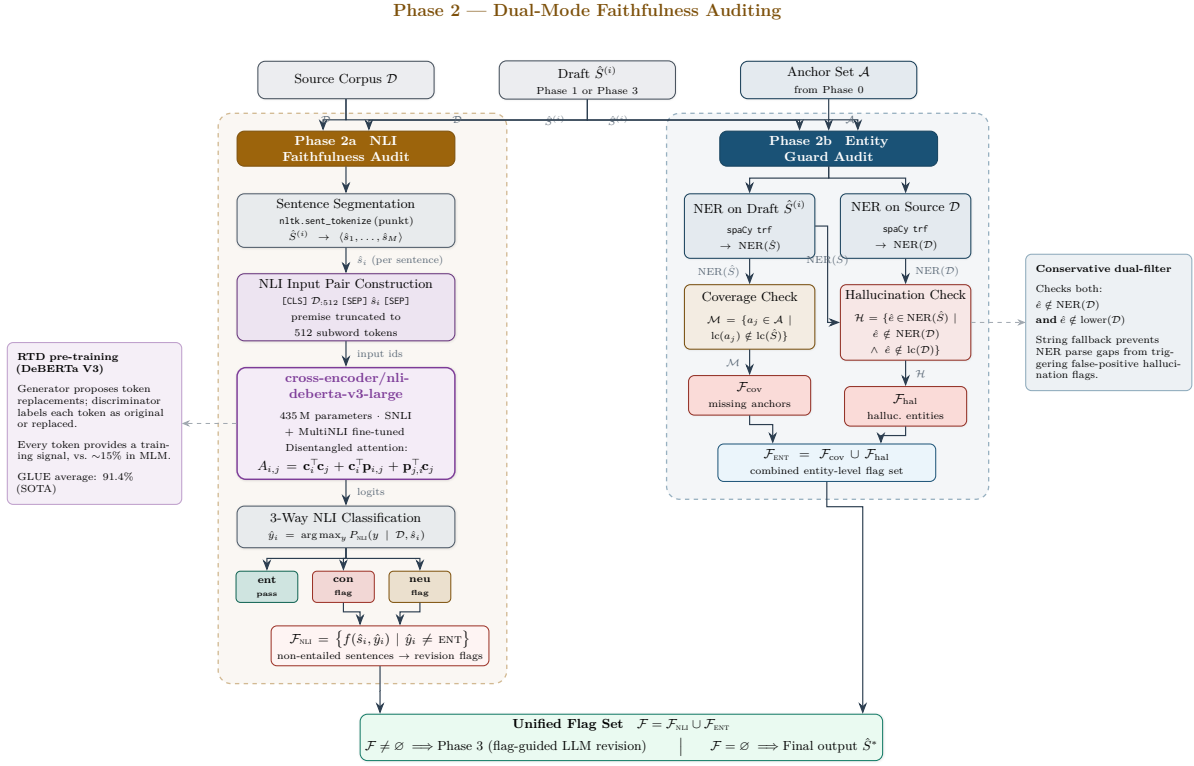


Figure 4: Phase 2 Architecture: Dual-Mode Faithfulness Auditing consisting of (a) NLI-based sentence validation and (b) Entity Guard coverage and hallucination checks.

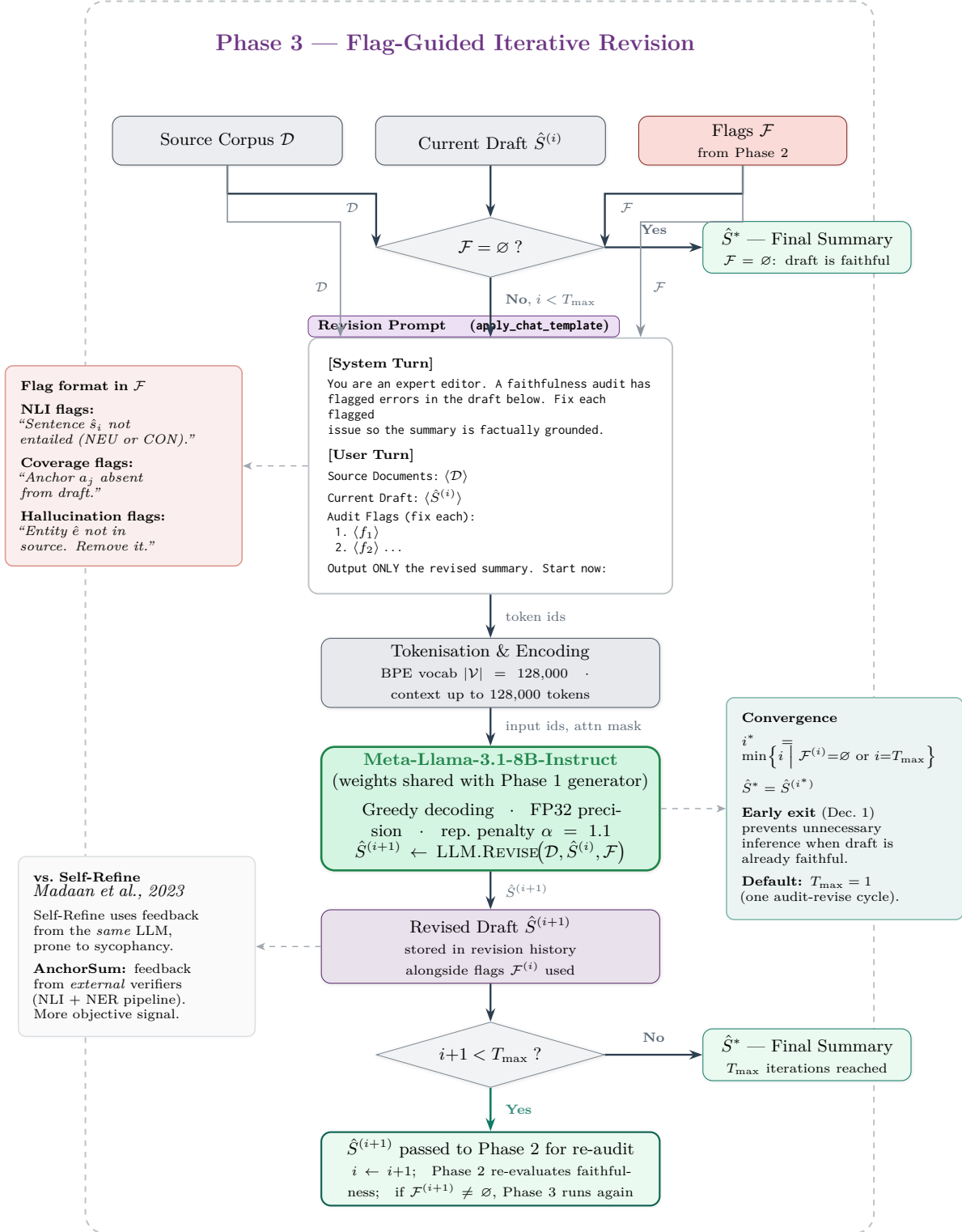


Figure 5: Phase 3 Architecture: Flag-guided iterative revision loop for error correction.